

Listening to History Rhymes: Pattern Recurrence and Return Predictability in Stock Markets

Abstract

Price-pattern studies specify in advance which patterns to look for. We let the data decide. We split U.S. daily equity returns, 1971 to 2025, into 1.3 million standardized 60-day segments and compare each with every other. The *Pattern Distance Ratio* (PDR) measures the crowding of real data around a segment against a random-walk benchmark. Recurrence is genuine and persistent: PDR's year-to-year correlation is 0.308, versus 0.011 for a placebo. Strongly recurring segments carry more reliable forecasts. A kernel ridge strategy in trajectory space earns gross annual alphas of 6% to 9.6%, and positive equal-weighted alphas in all twelve international markets.

Keywords: return predictability, matrix profile, time series analysis, machine learning, technical analysis.

History never repeats itself, but it does often rhyme.

— Mark Twain

1 Introduction

Traditional finance theory, notably the efficient market and random walk hypotheses, has long provided a foundational framework for understanding stock price dynamics. While these theories offer valuable first-order approximations, a large body of literature has documented numerous market anomalies (e.g., [Keim \(1983\)](#), [Ang et al. \(2006\)](#), [Banz \(1981\)](#), [Jegadeesh and Titman \(1993\)](#)). Researchers have often relied on ex-ante hypotheses regarding human behavior (e.g., [Stambaugh et al. \(2012\)](#), [Odean \(1998\)](#), [Shefrin and Statman \(1985\)](#)), market imperfections (e.g., [Duffie \(2010\)](#)), or predefined statistical rules such as those used in technical analysis (e.g., [Han et al. \(2013\)](#), [Zhu and Zhou \(2009\)](#), [Kwon and Kish \(2002\)](#)) to uncover regularities in stock market dynamics. Nevertheless, our understanding of the intricate mechanisms driving markets remains limited, given the vast number of participants, the complexity of market mechanisms, and the multifaceted interactions among market components in determining prices.

More broadly, most empirical approaches in finance organize return data primarily through a panel-data perspective, where the primitive observational unit is a firm-time observation and statistical analysis proceeds through cross-sectional regressions, factor models, rolling-window statistics, or characteristic-based forecasting. While these approaches have generated important insights, the underlying representation of the data remains fundamentally tied to the stock-date panel structure.

The most direct antecedent of our approach is [Lo et al. \(2000\)](#), who put the reading of price patterns on a rigorous footing: they use nonparametric kernel smoothing to automate the identification of chart formations and then ask, through the conditional distribution of subsequent returns, whether those formations carry information. Their program established that visual pattern recognition can be made an algorithmic, statistically testable object rather than a matter of chartist judgment. It also left two questions open, both of which follow from the fact that the

patterns themselves must still be named in advance. First, the search is confined to a dictionary of human-inherited shapes—head-and-shoulders, broadening tops, triangles—so a trajectory that recurs systematically but resembles no named formation is invisible to it, and one cannot ask which shapes recur, only whether the listed ones do. Second, because the object of study is a shape’s average conditional return, the analysis stops short of the properties that determine whether such structure is economically meaningful: whether a segment’s degree of recurrence is a stable characteristic that persists out of sample, and whether the structure survives the cost of trading on it. We take up the program from exactly this point. Replacing the predefined dictionary with an exhaustive comparison of every 60-day return trajectory against every other, we let the recurrent shapes emerge from the data; measuring each trajectory’s recurrence against a matched random-walk benchmark, we obtain a continuous characteristic whose persistence can be tested directly; and carrying the analysis through to a traded portfolio, we can say not only whether recurrence predicts returns but at what trading cost the portfolio’s alpha would be exhausted.

In this paper, we propose a complementary empirical perspective on stock return dynamics by reorganizing financial return data into a geometric space of standardized return trajectories. Rather than treating the stock-month observation as the primitive object of analysis, we partition the daily return history of U.S. equities into rolling 60-day return segments and embed these segments into a high-dimensional metric space, where similarity between trajectories is measured using z-normalized Euclidean distance. Under this representation, each observation becomes a return trajectory segment rather than a firm-date pair, and the geometry of the trajectory space itself becomes an object of empirical interest.

This representation opens a different class of empirical questions from those typically studied in traditional asset pricing and return predictability research. Instead of beginning with ex-ante hypotheses about particular economic mechanisms, firm characteristics, or predetermined chart patterns, our framework shifts attention toward the geometric organization of return trajectories themselves. This perspective naturally favors a more data-intensive and assumption-light approach, allowing large-scale structure to emerge from exhaustive comparison rather than from predefined conceptual classifications. It also enables the study of broader structural properties of

the trajectory space that are difficult to observe within conventional panel-data representations.

Once return trajectories are embedded into a geometric similarity space, many additional questions become natural objects of study, including the topology of trajectory clusters, the density distribution of regions in trajectory space, the temporal evolution of neighborhoods, and the interaction between market regimes and geometric structure. The present paper focuses primarily on recurrence and predictability as initial probes into this space, leaving broader structural analysis for future research.

Motivated by these questions, we draw on the key insight of the *Matrix Profile* (Yeh et al., 2016), originally developed for motif discovery in time-series data (Lin et al., 2002): that exhaustive pairwise comparison of subsequences using z-normalized Euclidean distance can uncover meaningful recurring structures even in data with extremely low signal-to-noise ratios—a defining characteristic of financial returns. While the original Matrix Profile framework was designed primarily for motif discovery within a single time series, we extend its underlying principle to the full cross-section of U.S. equities, enabling systematic comparison across approximately 1.3 million rolling return segments spanning 1971–2025.

Importantly, our contribution is not merely the application of a new computational technique to an existing finance question. Rather, the Matrix Profile serves as the enabling methodology for constructing a new representation of financial return data. In this representation, recurrence, neighborhood structure, clustering, and trajectory geometry become natural empirical objects of study. Our analysis of pattern recurrence through the Pattern Distance Ratio (PDR) should therefore be viewed as an initial exploration of this broader geometric framework rather than as a complete characterization of the space.

Using nearest-neighbor distances from exhaustive pairwise comparisons, we construct the Pattern Distance Ratio (PDR), which quantifies pattern recurrence relative to a random walk benchmark constructed by reshuffling returns across stocks within each calendar year. The PDR compares how closely a given return segment is matched in the real data versus the randomized benchmark, imposing no ex-ante pattern classifications or predefined chart structures. The normalization by the random-walk benchmark is essential rather than cosmetic. Raw nearest-

neighbor distances conflate genuine recurrence with the generic commonness of a segment’s shape and with the changing density of the comparison set over time: smooth, trend-like trajectories are close to many segments in any data—real or random—and neighbor distances shrink mechanically as the cross-section grows. By measuring each segment’s neighbor distances against a null constructed from the same returns, the same year, and the same search procedure, the PDR functions as a local density ratio—how much more crowded the real data are around a given trajectory than chance would imply—making recurrence comparable across shapes, stocks, and decades. Appendix D formalizes this interpretation in a latent-template model of recurring trajectories and derives the testable implications that organize our empirical analysis. This exhaustive, assumption-light search reveals that certain return trajectories recur with greater frequency and consistency than others. Throughout the paper, recurrence is therefore a statement about neighborhoods, not about exact repetition: no two 60-day trajectories are ever identical, and none need be. What recurs is the occupancy of regions in trajectory space—some neighborhoods attract far more segments than chance would imply. History rhymes without repeating.

Our empirical analysis yields several findings. First, we document that pattern recurrence is genuine and persistent. Under annual comparison blocks, the correlation between in-sample and out-of-sample PDR is 0.308 at a one-year distance and decays only to 0.270 at a three-year distance—the slow decay of a structural characteristic of return dynamics rather than a short-lived phenomenon. Lengthening the comparison blocks to two, three, and five years raises the correlation to 0.434, 0.486, and then 0.603, the precision gain from a larger comparison sample that our framework predicts.

A random-walk placebo, constructed by reshuffling returns across stocks within each year, produces a correlation of 0.011—roughly thirty times smaller than the one-year value of 0.308—ruling out the possibility that our results are artifacts of the distance metric, z-normalization procedure, or year-block comparison structure.

Transition matrices provide a complementary nonparametric perspective. Segments in the extreme PDR quintiles—both the most recurrent (Q1) and the most unique (Q5)—show the strongest tendency to remain in the same quintile over time, with Q5 retention probabilities

ranging from 40.8% to 54.6% across block lengths versus a null benchmark of 20%. The largely smooth off-diagonal decay indicates that movements within trajectory space are gradual rather than abrupt, consistent with PDR representing a slowly evolving structural characteristic of return trajectories.

We next establish a link between recurrence strength and return predictability. A sign-preservation analysis shows that strongly recurring segments maintain directional forecasts across time, with preservation rates of 51.4–54.1% in the four most recurrent PDR quintiles among segments with high neighbor-outcome scores—well above the independence-adjusted null of 50.2–50.4%. This systematic relationship between recurrence strength and predictability persistence suggests that the geometry of trajectory space captures economically meaningful regularities in stock return dynamics.

Finally, we translate these findings into a portfolio strategy. We map each return trajectory to its location in shape space—its Gaussian-kernel similarities to a grid of landmark trajectories drawn from the recurrence structure of Stage I—and predict future returns by ridge regression on this representation, in the spirit of the virtue-of-complexity literature (Kelly et al., 2024). Sorting firms each month into a long–short portfolio by predicted return, the strategy earns large gross risk-adjusted alpha—roughly 6% to 9.6% per year across the CAPM and Fama–French three-through six-factor models and equal- and value-weighted portfolios—that remains significant in the most recent subperiods. The strategy’s high turnover leaves it a break-even one-way trading cost of about 15 to 21 basis points. Finally, we transplant the engine, with every design choice frozen at its U.S. value, to twelve international equity markets: the equal-weighted long–short alpha is positive in all twelve, and a global country-average portfolio earns a significant return.

Technical analysis has consistently remained a preferred approach among investment practitioners. Menkhoff (2010) document the widespread reliance on technical analysis among fund managers across countries. On the academic side, while early research cast doubt on the statistical reliability of chartist approaches (e.g., Cowles (1933), Cowles (1944), Fama and Blume (1966)), subsequent studies have documented profitable trading rules and predictive structures (e.g., Brock et al. (1992), Bessembinder and Chan (1995), Lo et al. (2000)).

Our study aligns with this broad tradition in emphasizing time-series patterns in stock price movements. However, unlike conventional technical analysis, which typically relies on predetermined rules or handcrafted pattern definitions (e.g., [Sullivan et al. \(1999\)](#), [Bajgrowicz and Scaillet \(2012\)](#), [Brock et al. \(1992\)](#), [Neely et al. \(2014\)](#), [Goulding et al. \(2024\)](#)), our framework performs an exhaustive and assumption-light search over the space of return trajectories itself. In this sense, our methodology is fundamentally data-driven rather than rule-driven.

[Bajgrowicz and Scaillet \(2012\)](#) revisit the historical success of technical trading rules on the Dow Jones index, control for data snooping through the false discovery rate, show that an investor could not have selected the future best-performing rules *ex ante*, and find that the remaining performance is erased by low transaction costs. We find similarly low break-even trading costs from a very different starting point—a cross-sectional search over 1.3 million individual return trajectories rather than a rule menu applied to one index—and we regard the agreement as informative rather than redundant. The contribution here is not a tradable rule that survives where earlier ones failed; it is the object underneath. Whether a segment’s shape recurs turns out to be a measurable, continuous, and strongly persistent property of the data—one that survives a block-bootstrap null and a diversified neighbor rule, and that reappears in twelve international markets with every design choice frozen.

In recent years, advances in machine learning and computational methods have renewed academic interest in nonlinear structure extraction from financial data. For example, [Gu et al. \(2020\)](#) demonstrate the predictive power of tree-based and neural-network methods relative to traditional linear models. [Leippold et al. \(2022\)](#) document substantial predictability in Chinese equity markets using machine-learning techniques. [Jiang et al. \(2023\)](#) employ convolutional neural networks to extract predictive information from transformed price-chart images, while [Chen et al. \(2024\)](#) develop adversarial deep-learning methods for conditional asset pricing.

Our study differs from this literature in an important respect. Much of the existing machine-learning literature improves predictive performance within conventional panel-data representations of financial data. By contrast, our contribution lies primarily at the level of data representation itself. Rather than transforming returns into image form or relying on firm characteristics as

inputs, we organize financial return data directly into a geometric trajectory space and investigate the structural properties of that space.

Closest to our work is the contemporaneous study of [Chen et al. \(2026\)](#), who construct a high-dimensional economic-state vector from two centuries of newspaper text and show that the average market return following the most similar historical months forecasts the aggregate market—an explicit “history rhymes” design. Our paper shares the thesis but differs in every layer of implementation: we work in the cross-section of individual stocks rather than in aggregate market timing; our state object is the price trajectory itself rather than a news-derived description of the economic environment; and, before any prediction is attempted, our Stage I establishes that trajectory recurrence is a genuine, persistent, and placebo-robust property of the data—structure the prediction stage then exploits. The two papers also take opposite positions in an active methodological debate: [Chen et al. \(2026\)](#) deliberately keep the predictor simple—an equal-weighted average over nearest historical months—citing the caution of [Nagel \(2025\)](#) that high-dimensional kernel forecasts trained on short windows can degenerate into recency-weighted momentum, whereas our estimator retains the full high-dimensional similarity representation in the spirit of [Kelly et al. \(2024\)](#); Section 5.1 explains why the degeneracy identified by [Nagel \(2025\)](#) does not arise in our design. That two independent designs—one reading news text, the other reading price trajectories—converge on the same conclusion strengthens the case that recurring economic states leave predictable traces in returns.

While our approach retains the core principle of the Matrix Profile—exhaustive comparison of standardized subsequences via z-normalized Euclidean distance—our cross-sectional setting (approximately 1.3 million segments across 55 years) necessitates a departure from the standard FFT-based implementation. We instead compute distances directly using GPU-accelerated matrix operations, which accommodate our large-scale cross-sectional comparison structure while maintaining computational feasibility.

Our study makes several contributions to the asset pricing literature. First, inspired by the Matrix Profile’s principle of exhaustive subsequence comparison, we develop a scalable distance-based methodology for organizing and analyzing the geometric structure of stock re-

turn trajectories across the full cross-section of U.S. equities. Second, we introduce the Pattern Distance Ratio (PDR)—a self-normalized local density ratio that measures each segment’s recurrence against a shape- and year-specific random walk null—and provide evidence that recurrence in trajectory space is genuine and persistent. Third, we establish a direct link between recurrence strength and the persistence of return predictability. Finally, we demonstrate that information embedded in the geometric structure of return trajectories generates significant abnormal returns that remain unexplained by standard factor models.

The remainder of the paper is organized as follows. Section 2 describes the data and the Matrix Profile-inspired methodology for constructing the trajectory space and measuring recurrence. Section 3 presents the central empirical results on recurrence and persistence, together with robustness checks. Section 4 links recurrence strength to the persistence of return predictability. Section 5 develops the trajectory-space portfolio strategy and examines its performance, the role of the recurrence structure, the economic interpretation of the alpha, its trading costs, and its deployment in international markets. Section 6 concludes.

2 Data and Matrix Profile-Inspired Measurement of Price Pattern Recurrence

Figure 1 gives an overview of our method, from the raw data to the portfolio. This section describes Stage I of the method: how we identify recurring price patterns.

2.1 Data

We use three data sources: (i) daily stock market data from the Center for Research in Security Prices (CRSP), (ii) annual accounting data from Compustat, and (iii) daily and monthly Fama–French factor returns from Kenneth French’s data library, obtained through Wharton Research Data Services (WRDS). The CRSP data run from January 1970 to December 2025. The daily return sample begins in July 1971, because the book-to-market ratio used in the benchmark sorts below is based on each stock’s market equity at the end of the previous December. We describe

each step of the sample construction below.

We obtain daily returns, prices, and shares outstanding from CRSP for all common stocks (share codes 10 and 11) listed on the NYSE, AMEX, and Nasdaq (exchange codes 1, 2, and 3). Daily returns include delisting returns, following the standard procedure. Market equity is $ME_{j,t} = |P_{j,t}| \times SHROUT_{j,t}$,¹ summed across all share classes of each firm (*permco*). Following [Fama and French \(1993\)](#), we construct book equity from Compustat as $BE = SEQ + TXDITC - PS$, where preferred stock follows the standard order: redemption value, then liquidating value, then par value. We require $BE > 0$ and at least two years of Compustat data, to reduce backfill bias.

To construct characteristic-adjusted returns, we form 25 benchmark portfolios by sorting stocks independently into quintiles of market capitalization (size) and quintiles of the book-to-market equity ratio (BM), following [Fama and French \(1993\)](#).

We apply two filters to limit the influence of illiquid or very small stocks. These stocks may produce spurious patterns driven by microstructure noise rather than genuine price dynamics. First, we exclude a stock-year if the stock’s closing price at the end of June is below \$5. This filter removes penny stocks, whose returns can be dominated by bid-ask bounce and discrete price effects. Second, we exclude a stock-year if the stock’s market equity at the end of June is below the 10th percentile of NYSE firms’ market equity in the same month. This filter removes microcap stocks that may trade too little to produce reliable price patterns. We apply both filters every year: a stock excluded in one July-to-June period can re-enter the sample in a later period if it meets the criteria. The final sample covers 55 years (1971–2025) and contains about 1.3 million return segments.

2.2 Return Segments and Matrix Profile Distance Metrics

Our main objects of study are daily return segments of individual stocks, built with rolling windows. To keep the sample size manageable, the windows roll monthly and end on *month-end dates*. For each month-end trading day t , the *formation window* is the 60 trading days up to and

¹All variables and notations are defined in Table A1 in the Appendix.

including t . The return segments from these formation windows are the inputs to the Matrix Profile procedure in Stage I, where we identify recurring patterns in price dynamics.

In Stage II, described later in the paper, we study the investment implications of these patterns, using a return predictability framework. For this purpose, we define an *evaluation window*: the 20 trading days immediately after the *formation window*.

Return Segments

Each return segment is uniquely identified by the pair (i, t) , where i is the stock and t is the last day of the window. The corresponding 60-day return series is

$$\mathbf{R}_{i,t} = (r_{i,t-59}, r_{i,t-58}, \dots, r_{i,t}), \quad (1)$$

where $r_{i,s}$, for $s = t - 59, \dots, t$, is the daily return of stock i on trading day s . A missing daily return is set to zero when delisting returns are merged, and the next trading day is then dropped, so a segment needs a CRSP record on every day of its *formation window* and no segment contains a filled return. For brevity, we often index a segment by a single index n , with $n = (i, t)$. When needed, we write the underlying stock and date as $i(n)$ and $t(n)$, or simply i and t .

The Matrix Profile Framework and Distance Metric

Our distance-based approach builds on the Matrix Profile, a time-series technique introduced by [Yeh et al. \(2016\)](#) that extends earlier work on motif discovery ([Lin et al., 2002](#)). The Matrix Profile computes the z-normalized Euclidean distance between all pairs of subsequences in a time series to find recurring subsequences, or “motifs.” For two subsequences $T_{i,m}$ and $T_{j,m}$ of length m , the z-normalized Euclidean distance is

$$d_{i,j}^{MP} = \sqrt{2m \left(1 - \frac{QT_{i,j} - m \mu_i \mu_j}{m \sigma_i \sigma_j} \right)}, \quad (2)$$

where μ_i, μ_j and σ_i, σ_j are the means and standard deviations of the two subsequences, and $QT_{i,j}$ is their dot product. This distance equals the Euclidean distance between the two subsequences after each one is standardized to zero mean and unit variance. A small z-normalized distance means that two subsequences move in similar ways over time, after removing differences in level and scale.

We compute the z-normalized Euclidean distance d^{MP} in Equation (2) for the formation-window return segments. For 60-day segments $m = (i, t)$ and $n = (j, s)$, we write this distance as $d_{m,n}$ for brevity.² A smaller value of $d_{m,n}$ means that the two segments have similar dynamics over time, after accounting for differences in mean return and volatility. Throughout the paper, “L2 distance” refers to this z-normalized distance $d_{m,n}$.

The Matrix Profile algorithm uses the Fast Fourier Transform (FFT) to compare all pairs of subsequences within a single time series efficiently. Our setting is different. We compare return segments *across* the whole cross-section of stocks (about 1.3 million segments), which requires hundreds of billions of distance calculations, including comparisons with the random walk benchmark series. This cross-sectional structure does not fit the single-series setting for which FFT-based Matrix Profile algorithms are designed. We therefore compute the distances with GPU-accelerated matrix operations, using the CuPy library.³

2.3 Random Walk Benchmark

To test whether the similarities among real return segments reflect genuine recurrence or simple chance, we build a benchmark from randomly shuffled returns. We first state exactly which null hypothesis this benchmark represents. When we pool all returns within a calendar year

²Because both vectors are standardized, this distance is a monotonic function of the Pearson correlation. We standardize with the sample standard deviation (divisor 59), so $d_{m,n}^2 = 2 \times 59 \times (1 - \hat{\rho}_{m,n})$; this equals the Matrix Profile distance of Equation (2) up to the constant $\sqrt{59/60}$, which leaves every ranking and the PDR unchanged.

³All GPU computations are conducted on a system equipped with an AMD Ryzen 9 9950X3D processor and three NVIDIA GeForce GPUs (an RTX 5090, an RTX 5080, and an RTX 4090), across which the distance computations are distributed, running CUDA 12.8 and CuPy 13.6.0. Distance matrices are stored in single-precision floating-point (float32) format, retaining approximately 7 significant decimal digits of precision.

and reshuffle them, the resulting series have no time structure *and* no differences across stocks. Every synthetic stock-day is an i.i.d. draw from that year’s pooled return distribution. This null is deliberately stronger than the random-walk hypothesis itself. The random-walk hypothesis says nothing about whether returns share a common distribution. It fully allows heteroskedasticity, stochastic volatility, jumps, and differences across stocks. Real data can therefore, in principle, differ from the pooled-i.i.d. benchmark for reasons other than recurring trajectory shapes.

For this reason, the shuffle benchmark plays two separate roles in this paper, and we keep them apart. Its main role is as the *normalization device* inside the PDR. In this role, it controls for how common a segment’s shape is in general, for the size of the comparison set, and for the year’s marginal return distribution (Section 2.4). The pooled construction is exactly what this role requires. The role of *null hypothesis for recurrence*, in contrast, is left to a much stricter benchmark: the stationary block-bootstrap surrogate of Section 3.3. The surrogate keeps each stock’s own return distribution (including its skewness, tails, and jump frequency), its short-range dependence such as volatility clustering, and its factor comovement. The persistence results survive this surrogate test. Under the shuffle benchmark, if returns had no time or cross-sectional structure, the distances between standardized segments in the real data would be statistically indistinguishable from the distances to the shuffled series. Systematic differences indicate structure. The surrogate then shows how much of that structure is genuine shape recurrence rather than differences in return distributions or simple autocorrelation.

We build the random-walk benchmark by *cross-stock randomization*. Within each calendar year, we pool all daily returns of all stocks and randomly reassign them across all stock-date observations. Formally, let

$$\mathcal{R}_y = \{r_{j,t} : \text{stock } j \text{ is in the sample on trading day } t \text{ in calendar year } y\} \quad (3)$$

denote the set of all daily returns of all stocks in year y , and let $N_y = |\mathcal{R}_y|$ be the total number of returns that year. We draw a random permutation σ of $\{1, 2, \dots, N_y\}$ and assign the $\sigma(k)$ -th element of $\mathcal{R}_y$ to the k -th stock-date observation. In other words, each observation (j, t) in year

y receives a return drawn uniformly at random, without replacement, from the pool of all returns observed across all stocks in that year.

We repeat the shuffle four times to create four independent random-walk series. The first three series ($m = 1, 2, 3$) form the benchmark for the PDR denominator (Section 2.4), and the fourth series ($m = 4$) stands in for the real query segments in the placebo test.

2.4 Pattern Distance Ratio (PDR)

To measure pattern recurrence, we compare how similar each real return segment is to other real segments with how similar it is to segments from the random walk sample described above. The idea is simple: if a pattern truly recurs in real data, its nearest neighbors among real segments should be systematically closer (that is, have smaller L2 distances) than its nearest neighbors in the random-walk sample.

To organize the comparisons, and to match the persistence analysis in Section 3, we divide all return segments into annual blocks. Each segment belongs to the calendar year in which its *formation window* ends. Let $\mathcal{D}_y$ denote the set of all segments whose 60-day window ends in year y , for $y = 1971, 1972, \dots, 2025$.

For each annual block $\mathcal{D}_y$, we compute two types of distance matrices:

- **Within-block (in-sample):** L2 distances between every pair of segments (n, n') with $n, n' \in \mathcal{D}_y$, which give the distance matrix of $\mathcal{D}_y$ against $\mathcal{D}_y$.
- **Across-block (out-of-sample):** L2 distances between each segment $n \in \mathcal{D}_y$ and every segment $n' \in \mathcal{D}_{y+h}$ for lags $h = 1, 2, 3, 5$, which give the distance matrix of $\mathcal{D}_y$ against $\mathcal{D}_{y+h}$.

The persistence analysis uses two versions of the across-block comparison, and they should be kept apart. In the *block-length* specifications (the 1-Year, 2-Year, 3-Year, and 5-Year columns of Table 1, Panels A–C), we divide the sample into non-overlapping one-, two-, three-, or five-year blocks and compare each block with the next one. A longer block therefore increases two things at once: the time distance between the compared blocks *and* the size of the comparison sample.

In the *fixed-block, varying-lag* specification (Panel D of the same table), the blocks stay at one year and only the lag h changes. This design separates the effect of time distance from the effect of the size of the comparison sample.

We compute these distances four times in parallel. The first run uses real standardized returns for both segments and gives the Real-to-Real distances. Each of the other three runs uses one of the three benchmark random walk series ($m = 1, 2, 3$) described in Section 2.3 and gives Real-to-Random distances. In these runs, the query segment uses real returns and the candidate segment uses random returns.

Return segments from different stocks can have small L2 distances simply because they move with common systematic factors. To limit this effect, we never compare two segments that share the same formation window. Because segments roll monthly, this rule ensures that the formation windows of any two compared segments start at least one month apart.

For each real segment n , we find its $K = 100$ nearest neighbors by z-normalized Euclidean distance (Equation 2). We then compute two mean nearest-neighbor distances for segment n :

1. **Real-to-Real Distance** ($\bar{d}_n^{real \rightarrow real}$): Among all other real return segments (excluding those on the same calendar date as n), we find the K segments with the smallest L2 distances to n . The Real-to-Real Distance is the mean L2 distance to these nearest neighbors:

$$\bar{d}_n^{real \rightarrow real} = \frac{1}{K} \sum_{k=1}^K d_{n, v_k^{real}}, \quad (4)$$

where v_k^{real} is the k -th nearest neighbor of n among all other real segments. A smaller $\bar{d}_n^{real \rightarrow real}$ means closer matches in real data, which suggests frequent recurrence.

2. **Real-to-Random Distance** ($\bar{d}_n^{real \rightarrow rand}$): For each of the three random walk series $m \in \{1, 2, 3\}$, we compute the L2 distances between the real segment n and all segments of random series m , find the K nearest neighbors, and compute the mean distance

$\bar{d}_n^{real \rightarrow rand, m}$. The Real-to-Random Distance is the average across the three series:

$$\bar{d}_n^{real \rightarrow rand} = \frac{1}{3} \sum_{m=1}^3 \bar{d}_n^{real \rightarrow rand, m}, \quad \text{where} \quad \bar{d}_n^{real \rightarrow rand, m} = \frac{1}{K} \sum_{k=1}^K d_{n, v_k^{rand, m}}. \quad (5)$$

Here, $v_k^{rand, m}$ is the k -th nearest neighbor of n among the segments of random series m .

The **Pattern Distance Ratio (PDR)** of segment n is defined as

$$PDR_n = \frac{\bar{d}_n^{real \rightarrow real}}{\bar{d}_n^{real \rightarrow rand}}, \quad (6)$$

where $\bar{d}_n^{real \rightarrow real}$ (Equation 4) and $\bar{d}_n^{real \rightarrow rand}$ (Equation 5) are the mean distances from n to its K nearest neighbors in real and random data, respectively. The PDR compares how closely a segment is matched in the actual market with how closely it is matched in a random-walk benchmark.

If $PDR_n < 1$, the nearest neighbors of segment n are closer in real data than in random data, which indicates genuine pattern recurrence. If $PDR_n > 1$, segment n is less similar to other real segments than to random segments, which suggests that it is relatively unique in the real data.

Why Normalize by the Random-Walk Benchmark?

The denominator is a central part of the PDR. The raw nearest-neighbor distance $\bar{d}_n^{real \rightarrow real}$ alone is not a valid measure of recurrence, because the expected distance from a segment to its nearest neighbors changes automatically for reasons unrelated to genuine pattern repetition. There are three such reasons. First, even after z-normalization, different regions of the shape space have very different densities under the null. In other words, some shapes are common and others are rare, even in random data. Smooth, trend-like segments are close to many other segments, real or random, whereas jagged, high-frequency segments are generally far from every comparison segment. A ranking based on the numerator alone would therefore largely sort segments by how common their shape is in general, not by recurrence. Second, the size of the comparison set changes over time. The number of U.S. stocks, and hence the number of candidate neighbors

in each annual block, changes a great deal over the 55-year sample. Nearest-neighbor distances shrink automatically as the comparison set grows. Third, the marginal distribution of daily returns differs across years and market conditions, which shifts the whole distribution of distances.

The Real-to-Random distance in the denominator removes all three effects by construction. The random-walk segments come from the same pool of returns in the same calendar year, are standardized in the same way, and are searched with the same K nearest neighbors over a comparison set of the same size. So $\bar{d}_n^{real \rightarrow rand}$ measures how close segment n would be to its neighbors if returns had no time structure, given its own shape, the size of the comparison set, and the year’s return distribution. The ratio therefore has a natural reading as a *local density ratio*. Because nearest-neighbor distances are inversely related to local density (Loftsgaarden and Quesenberry, 1965), the PDR compares the density of real segments near segment n with the density of null segments at the same location in shape space. $PDR_n < 1$ means that the region of trajectory space around n is more crowded in real data than chance would imply. This indicates genuine recurrence, not merely that n has a common shape or comes from a year with many stocks. This self-normalization makes the PDR comparable across segments, stocks, years, and volatility regimes, which allows us to pool about 1.3 million segments from 1971–2025 into a single analysis.

Appendix D formalizes this interpretation in a latent-template mixture model of return trajectories. In the model, the probability limit of the PDR is an exact monotone function of the local density ratio. The raw neighbor distance, in contrast, is confounded by how common a shape is in general and by the size of the comparison set, and both effects cancel in the ratio. The model also delivers the testable implications we examine in the rest of the paper: a placebo PDR equal to one with zero persistence; two-sided detection, in which uniqueness is itself informative; persistence that strengthens as estimation noise falls; and a monotone link between recurrence strength and return predictability.

PDR Persistence

We test whether patterns with low PDR (strong recurrence) in one period also recur strongly in later periods. For each segment $n \in \mathcal{D}_y$, we have both an in-sample PDR (PDR_n^{in}) and an out-of-sample PDR ($PDR_n^{out,h}$). We compute the Pearson and Spearman rank correlations between PDR_n^{in} and $PDR_n^{out,h}$ across all segments in year y . Under the null hypothesis that stock returns follow a random walk and pattern recurrence is spurious, these correlations should be close to zero. A positive correlation indicates that the degree of pattern recurrence is a persistent characteristic that carries over to new data. We report both the pooled correlation (across all years) and the year-by-year correlations, to check whether the result is consistent over time.

PDR Quintile Transition Matrices

To study the stability of recurrence further, we build transition matrices based on PDR quintiles. For each block length h (one, two, three, or five years), we pool the patterns of all block pairs and sort them into five quintiles by their in-sample PDR_n^{in} (Q1 = lowest PDR, or strongest recurrence; Q5 = highest PDR, or weakest recurrence). We then sort the same patterns into quintiles by their out-of-sample $PDR_n^{out,h}$. The transition matrix $\mathbf{T}^h$ has entries

$$T_{q_1, q_2}^h = \Pr(\text{out-of-sample quintile} = q_2 \mid \text{in-sample quintile} = q_1), \quad (7)$$

estimated as the fraction of patterns in in-sample quintile q_1 that fall into out-of-sample quintile q_2 . Under the null hypothesis of no persistence, every entry of $\mathbf{T}^h$ should equal $1/5 = 0.20$. Diagonal entries above 0.20 indicate that patterns tend to stay in the same recurrence quintile over time.

3 Pattern Recurrence in Stock Returns

This section presents the central empirical finding of the paper. Using the exhaustive distance comparison of Section 2.2, which is inspired by the Matrix Profile, we show that stock price

patterns recur in a genuine and persistent way that is absent from random walk data. Section 3.1 shows that the degree of pattern recurrence, measured by the PDR, is a stable characteristic that carries over from in-sample to out-of-sample periods. Section 3.2 supports this finding with transition matrices that track how segments move across PDR quintiles over time. Section 3.3 shows that the persistence is robust to other benchmark constructions, neighbor definitions, and segment lengths, and that it extends to international equity markets.

3.1 Pattern Distance Ratio (PDR) Persistence

We begin by asking whether patterns recur persistently over time. Figure 2 and Table 1 report the persistence of the PDR under one-, two-, three-, and five-year comparison blocks, together with a random walk placebo test. In the placebo, a fourth independent random walk series (Section 2.3) takes the place of the real data in the PDR, so any apparent persistence can only come from the method, not from the data. Panel B of Table 1 reports the overall correlation between in-sample and out-of-sample PDR. The Pearson correlation rises with each longer block, from 0.308 with one-year blocks to 0.486 with three-year blocks and 0.603 with five-year blocks. The Spearman rank correlation rises from 0.251 to 0.444. These positive correlations mean that segments with low PDR (high recurrence) in one period tend to keep high recurrence in later periods. A longer block, however, increases the comparison sample at the same time as it increases the time distance (Section 2.4). Panel D separates the two effects by holding the comparison blocks at one year and changing only the horizon. The correlation is 0.308 one year apart, 0.275 two years apart, 0.270 three years apart, and 0.257 five years apart.

In sharp contrast, the random walk placebo gives a correlation of 0.011 (Spearman 0.008). This is roughly thirty times smaller than the one-year real-data value of 0.308, and it is visually flat in the decile plots of Figure 2. We compare magnitudes here, not statistical significance, and we do so on purpose. The placebo comes from a single random shuffle, so its persistence is itself a random variable. A significance test is also the wrong tool for this comparison. A tiny residual with no economic meaning can easily be “significant,” and failing to reject zero would not show that it is zero. What the placebo can show, and does show, is that the method

cannot create persistence as large as that in the real data. Panel C reports consistency across block pairs: the mean Spearman correlation across block pairs ranges from 0.233 (1-year) to 0.450 (5-year). The year-by-year Pearson correlation is positive in 100% of years at the one-year horizon (Figure 3). This universal positive persistence is strong evidence against the pooled-i.i.d. shuffle null. Section 3.3 shows that the persistence also survives a much stricter null that keeps differences in return distributions across stocks, volatility clustering, and factor comovement.

Figure 3 shows persistence over time, with gray shaded areas marking NBER recessions. It plots the Pearson correlation between in-sample and out-of-sample PDR for each year from 1971 to 2024. The correlation varies over time: persistence is especially strong in the 1990s and weaker in the 2000s and 2010s. Table 3 reports the directional counterpart of this persistence by PDR quintile: how often a segment keeps the sign of its neighbor-outcome score from in-sample to out-of-sample. Section 4 analyzes these results.

Several features of these results are worth discussing. First, Panel D shows that pattern recurrence decays slowly. Holding the comparison sample fixed, the correlation falls only from 0.308 at a one-year distance to 0.270 at three years and 0.257 at five years. It keeps almost ninety percent of its one-year value at three years and about 83 percent at five years. If recurrence came from short-lived factors such as temporary liquidity shocks or brief investor attention, the correlation would drop sharply with distance. The observed decay is mild. The steady *rise* across the block-length columns of Panel B has a different cause, and it does not mean that persistence grows with distance. Longer blocks contain more candidate neighbors. More candidates reduce the estimation noise in each segment’s nearest-neighbor distances. This automatically raises the correlation between the in-sample and out-of-sample estimates, exactly as Proposition 3(ii) of Appendix D predicts. The two panels together give a consistent picture: time distance alone lowers the correlation slowly (Panel D), while a larger comparison sample raises it (Panels B–C). Both patterns are what we would expect from a stable recurrence characteristic that is measured with noise.

Second, the contrast with the random walk placebo is sharp and economically informative. Placebo persistence is one to two orders of magnitude smaller, which confirms that the method

itself does not automatically create persistence of the size we document. Persistence of order 0.3 appears only when we apply the method to real market data. This rules out the possibility that our results are artifacts of the distance metric, the z-normalization procedure, or the year-block comparison design. In real data, the Pearson correlation is positive in 100% of individual years, with a mean of 0.317. This further shows that pattern recurrence is a widespread feature of stock markets, not limited to particular subperiods or market regimes.

3.2 Transition Matrices and PDR Stability

Table 2 reports transition matrices: the probability that a segment in PDR quintile q_1 (in-sample) moves to quintile q_2 (out-of-sample). The diagonal clearly dominates. The diagonal entries are the probabilities that segments stay in the same quintile. For 1-year blocks (Panel A), all five diagonal probabilities exceed 0.20, the value expected under a random walk. The extreme quintiles are well above the middle three: 0.269 (Q1) and 0.408 (Q5), compared with 0.229–0.248 for Q2–Q4. The diagonal becomes much stronger as the comparison blocks get longer. With 3-year blocks (Panel C), Q1 segments stay in Q1 with probability 0.310, and Q5 segments stay in Q5 with probability 0.485; with 5-year blocks (Panel D), the two probabilities are 0.345 and 0.546.

The two extreme quintiles, Q1 and Q5, deserve attention. Segments with the lowest PDR (Q1, highest recurrence) and segments with the highest PDR (Q5, lowest recurrence) both persist more than the middle quintiles. This suggests that both ends of the PDR distribution are stable pattern characteristics. By contrast, Panel E shows that the random walk benchmark gives nearly uniform transition probabilities around 0.20 in all cells, with no meaningful diagonal dominance. This clear difference confirms that the PDR captures genuine and persistent pattern characteristics, not transient noise.

The transition matrices offer several further insights. First, Q5 is more persistent on the diagonal (0.408–0.546) than Q1 (0.269–0.345). This difference is informative: segments that are unique, in the sense that they have no close matches in the data, tend to stay unique over time. This is consistent with the view that Q5 segments reflect idiosyncratic, firm-specific price

dynamics that are inherently hard for other stocks to replicate, whereas Q1 segments capture price dynamics shared by many stocks. Because both extremes persist more than the middle quintiles, the PDR distribution appears to have stable tails: segments tend to sort into “highly recurrent” or “highly unique” groups that persist across years.

Second, the off-diagonal entries show a clear gradient. In rows Q1–Q4, the probability of moving to a quintile falls steadily as that quintile gets farther from the diagonal. For example, in Panel A, a Q1 segment has a 23.4% chance of moving to Q2 but only a 13.3% chance of jumping to Q5. The only exception is the far end of the Q5 row, where a jump to Q1 is slightly more likely than a jump to Q2 (13.4% versus 11.3% in Panel A, with the same ordering in Panels B and C; in Panel D the two are 6.8% and 7.6%). This smooth gradient shows that PDR usually changes gradually rather than abruptly, which supports the view that the degree of pattern recurrence is a stable, slowly changing characteristic.

Third, the diagonal becomes stronger from the 1-year to the 5-year specification, just as the correlations do in Section 3.1. As with the correlations, this partly reflects the larger comparison sample of longer blocks (Section 2.4). The transition matrices also add a nonparametric confirmation: pattern recurrence is not only correlated over time in a linear sense but also shows genuine rank-order stability. Together, the correlations and the transition matrices show that the PDR is a meaningful and durable property of stock price patterns. It separates real market dynamics from data with no time structure at every horizon we examine.

3.3 Robustness

Table A2 shows that the persistence is robust to two natural concerns about how the PDR is built. The first concern is the benchmark. Our random walk null shuffles returns across stocks within each year, so it destroys each stock’s own time-series structure. Real 60-day segments could then look systematically closer to one another than to the benchmark merely because of trivial univariate autocorrelation, such as volatility clustering, rather than genuine shape recurrence. To address this concern, we rebuild the benchmark under a much stricter null: a stationary block bootstrap (mean block length of 10 trading days) applied to each stock’s six-factor (FF5+UMD)

residuals. This construction keeps the stock’s real factor exposures, and hence its systematic comovement and most of its own autocorrelation, and scrambles only the idiosyncratic part. The second concern is the neighbor set. A segment’s 100 nearest neighbors may be dominated by peers from the same period, industry, size group, and value group. Measured recurrence would then reflect common factor comovement rather than separate repetitions of a pattern. To rule this out, we use a stricter neighbor rule. We divide stocks into size $\times$ book-to-market $\times$ 48-industry $\times$ month cells and keep at most one neighbor per cell before taking the 100 nearest. This rule forces the neighbor set to be diverse across firms, sectors, and time.

Both alternatives leave the persistence result intact. Under the much stricter block-bootstrap surrogate, the mean year-by-year Pearson correlation between in-sample and out-of-sample PDR is 0.247 (Spearman 0.176). This is about 22% below the 0.317 baseline, but it is still large. Doubling the mean block length to 20 trading days does not weaken the persistence: the mean correlation is 0.262. With this block length, a 60-day window is built from about three blocks, so the null keeps within-window dependence up to monthly scales, including slow volatility dynamics. The persistence that survives the surrogate therefore does not come from slow volatility structure either. Under the stricter neighbor rule, persistence is, if anything, slightly stronger than the baseline (mean Pearson 0.350 versus 0.317). This confirms that the result is not an artifact of clustered, comoving neighbors. Both alternatives differ sharply from the random walk placebo, whose mean correlation of 0.014 is more than an order of magnitude smaller (Table A2).

We also carry the stricter null through to the transition analysis built on the PDR, not only to its persistence (Table A2, Panel B). When we sort segments on the surrogate-based PDR, the recurrent tail is, if anything, more stable than under the shuffle benchmark: the one-year Q1 retention probability rises from 0.269 to 0.362. The unique tail becomes less stable, falling from 0.408 to 0.296. This indicates that part of the baseline persistence of *uniqueness* reflects persistent stock-level differences in return distributions, which the surrogate absorbs into the null.

A third concern is that PDR persistence may simply restate the well-known persistence of volatility: volatile stocks trace unusual-looking paths, and volatility itself is highly persistent.

Table A3 shows that this is not the case. Adding the formation-window idiosyncratic volatility to the persistence regression leaves the coefficient on in-sample PDR almost unchanged, at 0.304 versus 0.317. Adding size, reversal, and the path's trend changes nothing further. Idiosyncratic volatility is itself significant, but it absorbs almost none of the persistence. Persistence is also present within every volatility tercile. The mean correlation is 0.215 among the least volatile segments, where it is positive in 49 of 54 year-pairs, and it rises to 0.389 among the most volatile segments. Persistence is therefore a property of the shape of the path, not of its scale. We conclude that PDR persistence reflects genuine idiosyncratic pattern recurrence rather than trivial autocorrelation, factor comovement, or volatility persistence.

The persistence is also robust to other segment lengths. With 30-day and 90-day segments in addition to the 60-day baseline, the Pearson correlation between in-sample and out-of-sample PDR ranges from 0.287 to 0.333 (Table A4). The corresponding sign-preservation results are reported together with the predictability results in Section 4.

Finally, the persistence is not limited to U.S. equities. We repeat the PDR construction in twelve international equity markets. The candidate pool is every Compustat Global market that could reach 5,000 segments in some year, and the depth rule explained below selects the twelve. The in-sample/out-of-sample correlation is positive in every market, and in at least 70 percent of the market-years within each market (Table A5). In six of the twelve markets, the mean correlation exceeds the U.S. mean of 0.317. The spread across markets is wide, from 0.49 in the United Kingdom to 0.09 in China. Section 5.5 returns to these twelve markets and applies the full prediction engine to them as an out-of-sample test.

The international data are daily prices from Compustat Global (*g_sec*d), cleaned as in Chaieb et al. (2021): common shares (issue type code 0) on each market's main exchange (the two largest exchanges in China, France, Germany, India, Japan, Korea, and Taiwan); a security-description screen that removes preferred shares, warrants and rights, bonds, funds, trusts, REITs, depositary receipts, certificates, and exchange-traded products; the most-traded issue of each firm; and prices from days with an actual trade. Because cross-listed securities and exchange-traded products carry quotes on local holidays, a date counts as a trading day only when the number of priced securities

is at least $0.8m - 2\sqrt{m}$, where m is the median count over the 21 nearest dates, and at least 20 percent of the securities priced that month; returns compound across the dates dropped. Daily total returns come from adjusted closing prices and total-return factors, and are set to missing on the day a security's quotation currency changes or when they exceed 100 percent in absolute value. June market equity, converted to U.S. dollars at that day's exchange rate, must exceed US\$10 million for a stock to enter the following July–June year.

The cross-sectional depth requirement needs a brief explanation. It determines which markets can be studied at all, and it is much stricter here than in the international asset-pricing literature. Studies that estimate factor exposures stock by stock need only a long enough time series for each stock. For example, [Chaieb et al. \(2021\)](#) cover 47 countries. In such studies, even a market with a few dozen listed firms provides usable observations. The PDR is a different kind of object. It is a *local density* statistic: the distance from a segment to its nearest neighbors *within the same market and year*, divided by the same quantity computed against a benchmark from that market and year. Its precision therefore depends on how densely the same-year cross-section fills the neighborhood of each segment, not on the length of any stock's history. In a thin market, the nearest neighbors are far away by construction, and both the numerator and the denominator are estimated with large error. As Proposition 3(ii) of Appendix D predicts, estimation noise then pushes the measured in-sample/out-of-sample correlation toward zero, even if structure is present. We therefore require each market-year to have at least 5,000 segments observed in both the in-sample and the out-of-sample block. We apply this rule without exception and keep markets with at least five qualifying market-years, the minimum for the cross-year test in Table A5. Twenty-five Compustat Global markets have enough listed common shares to reach 5,000 segments in some year (at least 417 issuers with a price in that year). Fourteen of them have at least one qualifying year; Indonesia (four years) and Turkey (three) fall short of five, and the other eleven, including Germany, have none. Sweden, with six, is the least deep of the twelve. Keeping a thin market would not bias the results toward our hypothesis, because noise weakens the correlation. It would, however, add uninformative rows measured with error, and we prefer to state the depth requirement openly rather than average thin markets into the results.

4 Recurrence Strength and Return Predictability

Having shown that price patterns recur persistently (Section 3), we now ask whether the degree of recurrence is linked to the persistence of return predictability. We measure predictability with the neighbor-outcome score: the T-statistic of the future Fama–French 25-adjusted returns of each segment’s nearest neighbors (defined formally in Appendix B). For each segment $n \in \mathcal{D}_y$, we compute in-sample statistics (PDR_n^{in}, T_n^{in}) from the within-block comparison of $\mathcal{D}_y$ with $\mathcal{D}_y$, and out-of-sample statistics ($PDR_n^{out,h}, T_n^{out,h}$) from the comparison of $\mathcal{D}_y$ with the next block. We use the one-, two-, three-, and five-year block lengths of Table 1 (Section 2.4).

Table 3 reports sign preservation rates: the share of segments whose directional prediction (the sign of T_n) is the same in-sample and out-of-sample. Among high- $|T|$ segments, the rates reach 51.4–54.1% in the four most recurrent quintiles, well above the independence-adjusted null of about 50.2–50.4%. The rates fall toward the null in the most novel quintile. For all four block lengths, sign preservation in Q1 (strongest recurrence) exceeds that in Q5 (weakest recurrence) by 1.6–2.4 percentage points. This result supports the link between recurrence strength and the reliability of predictive signals. The results are robust to other segment lengths and holding periods (Table A6). Across all six combinations of pattern window (30, 60, and 90 days) and holding period (20 and 30 days), high- $|T|$ sign preservation reaches 50.9–53.6% in the four most recurrent quintiles. It exceeds the independence-adjusted null in every such cell. It is also higher at the 20-day than at the 30-day holding period for every window, and it is lowest for the 90-day window.

Together with the persistence evidence of Section 3, these results indicate that the geometry of trajectory space captures economically meaningful regularities. In particular, the past outcomes of patterns in the four most recurrent PDR quintiles forecast future returns more reliably than those of the most novel quintile. Section 5 turns this segment-level link into a tradable long–short strategy and evaluates its risk-adjusted performance.

5 A Trajectory-Space Portfolio Strategy

We turn the pattern-based predictability of Section 4 into a tradable long–short strategy. The estimator directly implements the idea that location predicts return in the trajectory geometry: a trajectory’s expected future return is estimated from the realized outcomes of similar past trajectories. Section 5.1 describes how we build the strategy, and Section 5.2 reports its risk-adjusted performance. Section 5.3 asks whether the alpha is better read as a risk premium or as mispricing, and Section 5.4 measures the strategy’s turnover and break-even trading costs. Finally, Section 5.5 applies the engine, with every design choice frozen, to twelve international equity markets as a fully out-of-sample test.

5.1 Methodology

The estimator works entirely in the trajectory space of Stage I. After Stage I, the model never sees raw returns; it sees only the Stage I distance metric. The estimator is therefore best understood as an approximate kernel ridge regression in the trajectory geometry, that is, a distance-weighted average of similar past outcomes.

The construction has four steps. First, we fix a set of *landmark* trajectories drawn from the *recurrence grid* of Stage I. From the segments of the previous three years, we keep those whose neighbor-outcome score is at least the conventional cutoff in absolute value ($|t| \geq 1.96$). We then draw 410 of them from each decile of this pool’s PDR distribution. The grid is therefore anchored at the recurrence structures identified in Stage I and covers both the recurrent and the novel regions. The draw gives up to 4,100 landmarks per test year: 4,100 in forty-nine of the fifty-one test years, and 1,612 and 3,239 in the two earliest years, when the pool is thin. We take five independent draws and average their predictions by within-month rank, so that no single draw determines the signal. Second, we represent each 60-day standardized return trajectory by the vector of its Gaussian-kernel similarities to the landmark trajectories. This vector is a Nyström / random-feature approximation of the full kernel (Williams and Seeger, 2001; Rahimi and Recht, 2008). Third, we predict the FF25-adjusted forward return (Appendix B) by ridge regression

on this similarity vector. Before the regression, we residualize the target each month on the segment’s five-day and 21-day reversal characteristics, and the ridge target is the within-month percentile rank of that residual mapped through the inverse normal distribution (a rank-Gaussian target). We estimate the regression on an expanding, purged window, with a one-month embargo between the training sample and the prediction date. Fourth, following [Da et al. \(2014\)](#), we orthogonalize the fitted score each month to the same two characteristics, and the residual is the trading signal we use throughout. We keep the full high-dimensional similarity vector and regularize it with ridge, rather than selecting a few features, following the virtue-of-complexity logic for return prediction ([Kelly et al., 2024](#)).

One point about the landmarks is essential for reading the results that follow. The landmarks are a computational basis, not the source of the forecast. The information the estimator uses lies entirely in the training pairs of trajectories and realized outcomes. The landmark set only fixes the finite-dimensional basis in which the kernel regression is expressed. Under the framework of Appendix D, any landmark set that covers the dense regions of trajectory space spans nearly the same space of smooth functions of location (Proposition 4(iii) and Remark 2). The estimator’s performance should therefore be nearly the same however the landmarks are chosen. In particular, landmarks drawn at random from the pool of real trajectories are not a recurrence-free alternative. Random draws from real data land in the dense regions of shape space in proportion to their density, and that density structure is exactly the recurrence documented in Section 3. We use the recurrence grid because it is a natural, pre-specified way to fix the basis. The evidence linking recurrence to predictability, however, does not depend on the landmark choice. It rests on the landmark-independent results of Sections 3 and 4.

[Nagel \(2025\)](#) warns that the virtue-of-complexity logic can be misleading. When a high-dimensional kernel regression is trained on a short rolling window, similarity in the predictor space mainly reflects closeness in time. The forecast then reduces to a recency-weighted average of recent returns: an implicit momentum strategy disguised as complexity. Two features of our design rule out this problem. First, we forecast the cross-section, not the aggregate market. Stocks are ranked within each month, so any common recency component cancels out of the long–short

position. Second, the training window is expanding, not short and rolling: it spans decades and puts equal weight on all history. The predictive map from trajectory shape to future return is stable over long periods. It is not a repackaging of recent returns.

At the end of each month, we sort stocks into quintiles by the signal. We trade a long–short portfolio that buys the top quintile (Q5) and sells the bottom quintile (Q1), with both equal and value weights, over the out-of-sample period 1975–2025.

5.2 Performance

Table 4 reports the strategy’s long–short alphas under several factor models. The alpha is large and highly significant under every model, for both equal- and value-weighted portfolios (FF6 alphas of 8.39% and 5.92% per year, respectively). The engine reads a sixty-day price path whose final days are the most recent week, so a natural question is how much of the alpha is short-horizon reversal (Jegadeesh, 1990; Lehmann, 1990). The alpha survives adding French’s published *monthly* reversal factor (FF6 + ST-Rev: 8.54% equal-weighted and 5.81% value-weighted). It also survives when we add a *weekly* reversal factor (WST-Rev) on top of the monthly factor: the alpha is then 7.30% equal-weighted and 6.33% value-weighted. We build WST-Rev inside our own universe using the strategy’s own rule.

The strongest evidence for the recent period is out-of-sample by construction. When we apply the engine with every design choice frozen, its global country-average portfolio earns a significant raw return across twelve international markets over 2001–2025 (Section 5.5).

A remaining concern is that we chose the estimator after looking at the data, so the reported *t*-statistics may overstate the evidence. Here the label “out-of-sample” needs care, because it can mean two different things. In the statistical sense (no forecast uses information realized after its formation date), the design is strictly recursive. The training windows expand, are purged, and have an embargo, and the landmarks are drawn point-in-time. In the stronger, design-level sense (no specification choice was made with knowledge of test-period performance), the relevant question is how much room such choices leave. Several apparent degrees of freedom are not free. The kernel bandwidth is set by the median heuristic, and the ridge penalty is

set by generalized cross-validation. These are data-driven rules fixed in advance, not tuned to performance. Section 5.5 applies the same specification, unchanged, to twelve international markets.

A further question is what the signal is, not whether it is real. Because the engine reads price paths, it may simply repackage technical rules and characteristics already known to predict returns. Table 6 tests this directly, in the format of Han et al. (2016). Every regressor is winsorized and standardized within the month, so slopes are comparable across variables, and we run the regressions both equal- and value-weighted. Moving-average rules built from the same 60-day window do predict returns in our sample, and so do reversal, momentum, idiosyncratic volatility, size, and liquidity. Yet when all of them are included together, the signal’s equal-weighted Fama–MacBeth coefficient is 0.152, and its t -statistic is still 8.2. Adding the weekly reversal characteristic does not reduce the coefficient (0.182, $t = 8.3$). In value-weighted regressions (Table A7), the coefficient is 0.076 ($t = 3.1$) with the technical rules and standard predictors included, so the signal’s incremental content is concentrated in smaller stocks.

The strategy rests on the premise that a pattern’s future return is predicted by the realized outcomes of similar past patterns. Because the landmark basis is interchangeable (Section 5.1), the link between the recurrence structure of Stage I and the strategy’s performance must be shown with evidence that does not depend on the landmark choice. This paper provides one such piece of evidence: the Stage I gradient linking recurrence strength to the persistence of neighbor predictability (Table 3).

5.3 Interpretation: Risk Premium or Mispricing?

The predictability could in principle be compensation for a priced risk, or a mispricing that arbitrage removes over time. The two readings make different empirical predictions. A risk premium should be roughly unaffected by the cost of arbitrage and need not weaken as arbitrage capital enters. A mispricing, in contrast, should concentrate in securities that are hard to arbitrage and should weaken as arbitrage capital competes it away (Shleifer and Vishny, 1997; McLean and Pontiff, 2016; Stambaugh et al., 2015). A raw long–short alpha is automatically larger

where returns are more dispersed, so we use two scale-free measures that separate signal quality from dispersion: the rank information coefficient and the Sharpe ratio (Table 5). One piece of evidence favors the mispricing reading. The predictability is somewhat stronger in small stocks. The small-cap tercile has a higher equal-weighted Sharpe ratio than the large-cap tercile in every window, and a modestly higher rank-IC in the full sample and after 2000 (Table 5: rank-IC 0.028 versus 0.021 over the full sample and 0.025 versus 0.023 after 2000; equal-weighted Sharpe ratios 1.51 versus 0.67 and 1.25 versus 0.65). The rank-IC difference is modest even though the Sharpe difference is not. We therefore read the size pattern as consistent with, but not decisive evidence for, limits to arbitrage that bind most among small firms (Shleifer and Vishny, 1997). This pattern fits a mispricing story more naturally than a stable risk premium. We stress, however, that this evidence is suggestive, not decisive. Because of the joint-hypothesis problem (Fama, 1970), a risk-adjusted alpha is always a joint test of market efficiency and of the assumed factor model. We cannot rule out that the predictability compensates for exposure to a priced risk that our factor models miss. Our evidence is consistent with limits to arbitrage, but it does not by itself exclude a risk-based explanation. We therefore read the results as favoring a mispricing interpretation, without claiming to settle the question.

Under this reading, one question remains. A significant alpha survives in large, liquid stocks after 2000 (Table 7, Panel B). After 2009, it is significant at the 1% level only in the value-weighted portfolio. Which limit to arbitrage preserves it? The natural candidate is the cost of trading it, to which we now turn.

5.4 Implementability and the Cost of Arbitrage

The alphas reported above are gross of trading costs, and the signal changes quickly: under monthly rebalancing, the strategy replaces 82–84% of each leg every month. This subsection asks what survives realistic transaction costs, using the accounting standard of the literature on anomaly trading costs (Novy-Marx and Velikov, 2016; Chen and Velikov, 2023). We compute turnover after letting the previous month’s weights drift with their realized holding-period returns, so that passive price changes are not counted as trading.

Table 7 reports the results in the form used by Grundy and Martin (2001): what the strategy earns, how much it trades, and the trading cost that would exhaust its earnings. We report the break-even cost rather than a net-of-cost alpha for two reasons. First, the break-even cost does not depend on how well spreads can be measured, because it uses only the alpha and the turnover. Second, what decides implementability is the reader’s own execution cost, not ours (Bajgrowicz and Scaillet, 2012). At this turnover, the alphas of 8.4% and 5.9% per year are exhausted by a one-way cost of 21.3 basis points equal-weighted and 14.6 value-weighted. Restricting both legs to the largest NYSE size tercile lowers this hurdle rather than raising it, because the gross alpha falls while the turnover barely changes. The hurdle then becomes 11–18 basis points under monthly rebalancing. The last column applies the stricter of Grundy and Martin’s two conventions: the cost at which the profit can no longer be distinguished from zero, rather than the cost at which it vanishes. This cost is naturally lower, 9–17 basis points for the monthly-rebalanced portfolio. It falls to 7.3 basis points among the largest firms after 2009 value-weighted, and to 2.3 equal-weighted. In those cells, trading costs would erase the statistical evidence before they erase the profit.

Our own estimates of this kind, the high–low spreads that serve as a control in Table 6, put the holdings-weighted one-way half-spread at about 47 basis points equal-weighted and 34 value-weighted, above every break-even cost in Table 7; at those spreads the alpha would not be captured. Daily high–low estimators overstate effective spreads for liquid stocks after decimalization (Chen and Velikov, 2023, Figure 2), and realized institutional costs are lower than academic spread estimates (Frazzini et al., 2018). A trader whose costs are in the single digits of basis points would clear the 13–17 basis-point hurdle of the post-2009 large-cap portfolio (the $\alpha = 0$ column of Table 7). In the modern large-cap universe, whether the strategy is profitable after costs depends on execution quality.

The predictability documented in this paper should be read as evidence about how prices form, not as a claim of implementable arbitrage profits.

5.5 Out-of-Sample Evidence from International Markets

The most demanding test of a return predictor estimated on fifty years of U.S. data is to apply its specification, unchanged, in markets it has never seen. We move the engine to the twelve international equity markets of Table A5, where Section 3.3 showed that the recurrence structure replicates. We use daily Compustat Global data, cleaned as described there. That table reports each market’s segments, stocks per month, and out-of-sample window. These screens play the role that the Ince and Porter (2006) and Hou et al. (2011) filters play in Datastream-based international studies. Most of the problems those filters target, such as non-common securities and padded post-delisting returns, do not arise in Compustat Global once the issue-type screen is applied. The file also retains inactive securities, so the sample is free of survivorship bias. Returns are in local currency. Because both legs of each long–short portfolio are in the same currency, exchange-rate movements affect the long–short return only at second order.

The test design follows the conventions of the literature that transplants anomalies to international markets (McLean et al., 2009; Watanabe et al., 2013; Jacobs and Müller, 2020): portfolios formed within each market, regional factor adjustment (Fama and French, 2012), and pooled cross-country regressions on within-country ranks. Every design choice is frozen at its U.S. value: the recurrence-grid landmark rule, the median-bandwidth kernel, the ridge-penalty selection rule, the purged expanding window, the rank-based target, and the portfolio rules. To be precise, “frozen” means that every numerical setting and every estimation rule is kept at its U.S. value; the objects they produce (each market’s landmark grid, bandwidth, penalty, and regression coefficients) are re-estimated from that market’s own data, exactly as a researcher applying the U.S. specification abroad would do. Three details differ from the U.S. engine because the international data and the Stage I outputs available abroad require them. Abroad, no FF25 benchmark exists, so the prediction target is the 20-day forward return in excess of the equal-weighted return of the other stocks in the same market; the reversal residualization and the rank transform are unchanged. The $|T|$ screen for the landmark pool uses each segment’s Stage I neighbor statistics computed within its own calendar-year block rather than against the three prior years, and when

fewer than 1,000 segments in the trailing three years pass the screen, the engine draws 4,100 landmarks uniformly at random from the trailing pool instead of skipping the year (Hong Kong 2003 and Sweden 2009–2012; 60 of the 3,401 market-months in Table 8). Code and a replication package are available from the authors.

Out-of-sample predictions begin between 1993 and 2009, depending on when a market’s cross-section first supports the minimum training sample of 20,000 segments (the U.S. engine uses 50,000; its cross-section is several times larger), and run through December 2025. Within each market, we evaluate the monthly quintile long–short portfolio against the corresponding regional five-factor-plus-momentum model (Fama and French, 2017). The factor series come from Kenneth French’s data library. We use the Europe, Japan, and Asia Pacific ex Japan series for the developed markets, and the library’s Emerging Markets series for China, India, South Korea, Malaysia, Taiwan, and Thailand. We benchmark regional portfolios against local rather than global factors, following Fama and French (2012), who show that global models fail to price regional portfolios. The emerging-market series comes from the same library and is built by the same procedure, but it lies outside the developed-market samples of Fama and French (2012) and Fama and French (2017). No international short-term reversal factor is available.

Table 8 reports the results. Three layers of evidence emerge. First, the signs: under regional factor adjustment, the equal-weighted long–short alpha is positive in all twelve markets. The equal-weighted alpha is individually significant at the 5 percent level in eight of the twelve markets. The four exceptions, China, South Korea, Thailand, and Taiwan, are positive but not significantly different from zero (t between 0.7 and 1.7). Second, the global series: we construct it as the *country-average portfolio* of Chui et al. (2010), which puts equal weight on each country’s own long–short portfolio. (Asness et al. 2013 aggregate in the same way but weight each market by the inverse of the return volatility of that market’s passive benchmark index.) We follow their terminology on purpose: in Chui et al. (2010), the *composite* portfolio is the contrasting object that pools stocks across markets. One element of our implementation is our own rather than theirs: we require at least five active markets in a month, while they require two. With equal weights across countries, the global portfolio earns 6.19% per year when stocks are equal-weighted within

each country ($t = 11.1$) and 4.92% when they are value-weighted ($t = 5.7$) over 2001–2025. Following Chui et al. (2010) and Jacobs and Müller (2020), we treat these raw figures as the main statistic for the aggregate. French’s Developed factors do not span the six markets in our set that belong to his Emerging universe, so a regional-factor adjustment of the aggregate is less well defined than it is market by market. Splitting the portfolio so that each half faces a model that does span it settles the question directly. The six markets in French’s Developed universe earn 8.12% per year against the Developed factors ($t = 10.4$) equal-weighted and 6.91% ($t = 5.3$) value-weighted. The six in his Emerging universe earn 5.92% against the Emerging factors ($t = 6.9$) equal-weighted and 4.51% ($t = 3.5$) value-weighted. The result therefore holds on both sides of the split under both weightings.

Third, Panel B reports a pooled cross-country Fama–MacBeth regression. We express the signal and the controls as within-country-month percentile ranks. The signal has a slope of 0.63% per 20-day holding period ($t = 9.8$) on its own, and 0.57% ($t = 10.0$) after controlling for size, book-to-market, prior-month reversal, and 12–2 momentum. This is the control set of McLean et al. (2009) and Watanabe et al. (2013), the two studies that run the same kind of pooled stock-level regression across markets. Because the engine reads a price path that ends with the most recent week, the third column also controls for the last-week return. The signal’s slope barely moves: 0.58% ($t = 10.1$), and it is significant in every column of Panel B. Book-to-market is itself priced in this sample (0.66, $t = 3.2$), so the value effect is present, and the signal simply does not overlap with it. The international predictability is therefore not a repackaging of the characteristics controlled for in Panel B.

Two features of the international sample should be noted. First, beyond the Chaieb et al. (2021) screens, the sample applies a \$10 million market-capitalization floor. This floor is our own addition, not part of their protocol. It plays the role that the NYSE size percentile and the \$5 price screen play in the U.S. sample, but it is an absolute threshold and a much weaker one. As a result, the large equal-weighted magnitudes in some markets (Australia, India, Sweden) are concentrated in small firms. Second, the out-of-sample windows vary widely in length, depending on when a market’s cross-section first supports the minimum training sample. They

range from seventeen calendar years for Sweden to thirty-three for Japan, or 204 to 396 months with a populated long–short portfolio. The return magnitudes should be read with this in mind. The returns are gross of trading costs, and we make no implementability claim abroad.

6 Conclusion

This paper is inspired by the Matrix Profile, a time-series technique that finds recurring motifs by comparing all pairs of subsequences. We develop a distance-based method to study whether price patterns in stock returns truly recur and whether such recurrence predicts future returns. We extend the core idea of the Matrix Profile from its original single-series setting to the full cross-section of U.S. stocks from 1971 to 2025. We use GPU-accelerated computation to compare about 1.3 million 60-day return segments. Unlike traditional technical analysis, which tests predefined chart patterns, our approach searches for all recurring patterns without prior assumptions about their shape or characteristics.

We introduce the Pattern Distance Ratio (PDR), which compares how closely a return segment is matched in real data with how closely it is matched in a random walk benchmark. Dividing by the random-walk benchmark is what turns a raw neighbor distance into a measure of recurrence. It removes the automatic effects of how common a segment’s shape is in general and of the changing density of the comparison set. As a result, the PDR works as a local density ratio that is comparable across shapes, stocks, and decades. Segments with a PDR below 1 recur more than chance would imply.

Our empirical analysis yields several main findings. First, pattern recurrence is strongly persistent over time. With the comparison blocks fixed at one year, the Pearson correlation between in-sample and out-of-sample PDR is 0.308 at a one-year distance and falls only to 0.270 at a three-year distance. Longer comparison blocks of three and five years raise the correlation to 0.486 and 0.603, because a larger comparison sample gives more precise estimates. In contrast, the random walk placebo gives a correlation of 0.011, roughly thirty times smaller than the one-year correlation of 0.308. This confirms that the method cannot create persistence as large as

that in real data. The persistence survives a much stricter block-bootstrap null and a diversified neighbor rule. It also extends beyond the U.S. to twelve international equity markets, where the same construction shows positive persistence.

Second, segments that recur more strongly show more persistent return predictability. The sign preservation analysis shows that segments keep their directional predictions from in-sample to out-of-sample periods. High- $|T|$ segments reach preservation rates of 51.4–54.1% in the four most recurrent PDR quintiles, and the rates fall toward the null in the most novel quintile.

Third, pattern-based predictability translates into economically large portfolio returns. We map each return trajectory to its location in shape space, given by its Gaussian-kernel similarities to a grid of landmark trajectories, and we predict future returns by ridge regression on this representation. The resulting long–short strategy earns large gross risk-adjusted alphas of roughly 6% to 9.6% per year across the CAPM and the Fama–French three- through six-factor models and across equal- and value-weighted portfolios. The alphas remain significant in the most recent subperiods. Because of its high turnover, the strategy has a break-even one-way cost of about 15 to 21 basis points. The predictability also survives the most demanding external-validity test we can construct. In this test, we freeze every design choice and move the strategy to twelve international markets. The strategy’s equal-weighted long–short alpha is positive in all twelve, and a country-average portfolio across these markets earns a significant return.

Our findings contribute to the asset pricing literature in several ways. First, we provide comprehensive evidence that stock price patterns truly recur, that this recurrence persists over time, and that returns without time structure cannot explain it. Under the block-bootstrap surrogate, differences in return distributions across stocks, volatility clustering, and factor comovement cannot explain it either. The contrast between the real data and these benchmarks shows that the recurrence reflects systematic market dynamics, not statistical artifacts. Second, we establish a direct link between the strength of pattern recurrence (measured by the PDR) and the persistence of return predictability. Patterns that recur more strongly have more reliable future performance, which suggests that pattern recurrence captures an economically meaningful dimension of stock return dynamics. Third, we show that pattern-based predictability generates significant abnormal

returns. Standard risk factors, including the market, size, value, momentum, profitability, and investment factors, do not explain these returns.

Both pattern recurrence and return predictability persist over long horizons. This indicates that standard pricing models do not fully absorb the information in price trajectories, either because the information compensates for risks those models omit or because frictions limit its arbitrage.

Our analysis suggests several directions for future research. First, studying the behavioral or institutional mechanisms that generate persistent price patterns would help explain why pattern recurrence is linked to the persistence of return predictability. Second, extending the analysis to other asset classes would test whether our findings hold beyond equities. Third, examining how pattern-based predictability varies with market conditions, liquidity, and information environments could show when and why patterns recur more strongly. Finally, the broader structural properties of the trajectory space itself remain largely unexplored. They include the topology of trajectory clusters, the density dynamics of regions in shape space, and their interaction with market regimes. These properties form a natural research program, and this paper is its first step.

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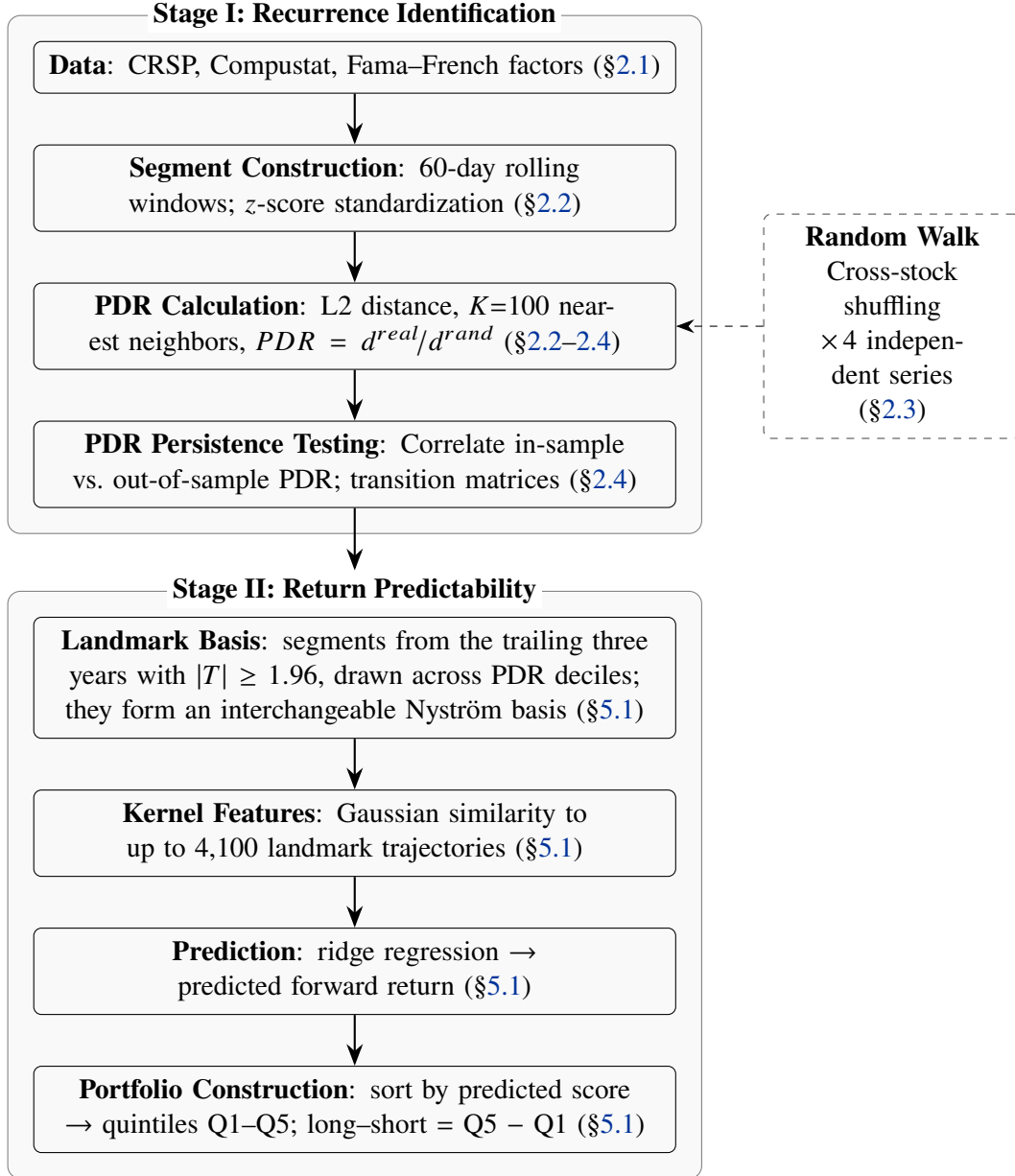


Figure 1: Overview of the methodology. The procedure has two stages. *Stage I (Recurrence Identification)* constructs return segments, computes the Pattern Distance Ratio (PDR) relative to a random walk benchmark, and tests whether recurrence persists over time. *Stage II (Return Predictability)* first fixes a basis of landmark trajectories drawn from the Stage I recurrence grid (Section 5.1). It then represents each trajectory by its kernel similarities to the landmarks and forms portfolios from ridge predictions. The dashed box marks the random walk benchmark (Section 2.3). Section references appear in parentheses.

Alt text: Flowchart of the two-stage procedure. Stage I runs downward through four boxes: data, segment construction, PDR calculation, and PDR persistence testing. A dashed random-walk box to the right feeds the PDR calculation. An arrow leads to Stage II, which runs through landmark basis, kernel features, ridge prediction, and portfolio construction.

Figure 2: PDR Persistence Comparison

This figure shows the persistence of the Pattern Distance Ratio (PDR) across comparison-block lengths. The figure is a two-row grid. Each column is one specification: the 1-Year, 2-Year, 3-Year, and 5-Year block lengths and a Random Walk placebo test. In the top row, each panel is a scatter plot of out-of-sample PDR against in-sample PDR, with one point per return segment. A random subsample of 3,000 segments is plotted for legibility. The dashed 45-degree line marks perfect persistence, and the dotted gray lines mark $PDR = 1$. Each panel reports the pooled Pearson correlation r . In the bottom row, each panel sorts segments into deciles of in-sample PDR. It then plots the mean in-sample PDR (circles) and the mean out-of-sample PDR (squares) within each decile. The shaded band shows the gap between the two. For the real-data specifications, the in-sample and out-of-sample decile curves track each other, and the correlation rises as the comparison block lengthens. Panel D of Table 1 separates this comparison-sample effect from the effect of temporal distance. Under the Random Walk benchmark, the scatter is uncorrelated and the out-of-sample decile curve is flat, as expected. The in-sample curve rises by construction, since deciles are sorted on it. All analyses use data from 1971–2025.

Alt text: A grid of two rows and five columns. The top row shows scatter plots of out-of-sample against in-sample PDR for one-, two-, three-, and five-year comparison blocks and a random-walk placebo. The real-data clouds tilt upward along the 45-degree line, with correlations of 0.308, 0.434, 0.486, and 0.603, while the placebo cloud is round with a correlation of 0.011. The bottom row plots mean in-sample and out-of-sample PDR by in-sample decile. For real data both lines rise together from about 0.98 to 1.05. For the placebo the in-sample line rises but the out-of-sample line is flat at 1.00.

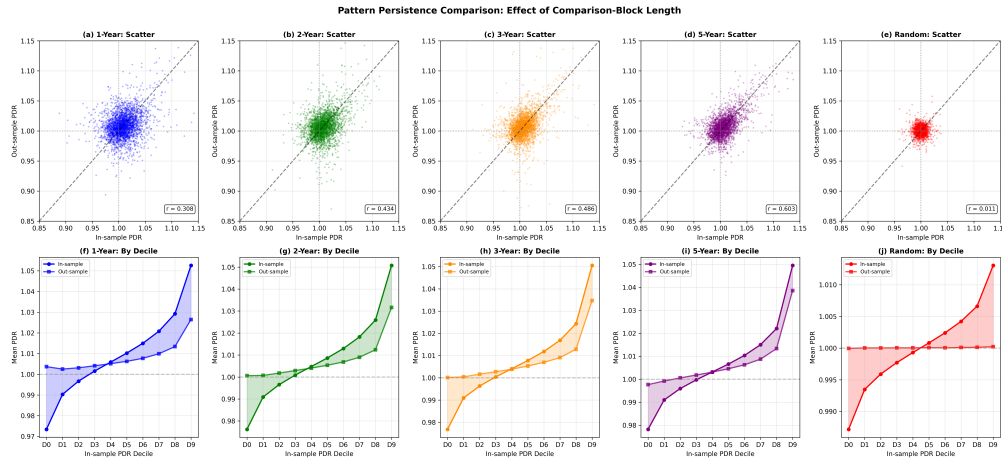


Figure 3: Yearly Correlation Trends

This figure plots the Pearson correlation between in-sample and out-of-sample Pattern Distance Ratio (PDR) for each year from 1971 to 2024. Gray shaded areas mark NBER recession periods. The horizontal dotted line marks the full-sample mean of 0.317. The series varies notably over time. Persistence is particularly strong during the 1990s and weaker in the 2000s and 2010s. Table 3 reports the directional counterpart of this persistence by PDR quintile: the sign preservation of the neighbor-outcome score.

Alt text: Line chart of the yearly correlation between in-sample and out-of-sample PDR from 1971 to 2024. The line stays above zero in every year. It rises from about 0.2 in the 1970s and 1980s to a peak above 0.6 in the mid-1990s, falls to a low near 0.05 in 2009, and fluctuates between 0.1 and 0.35 afterwards. Gray bars mark NBER recessions and a dotted horizontal line marks the mean of 0.317.

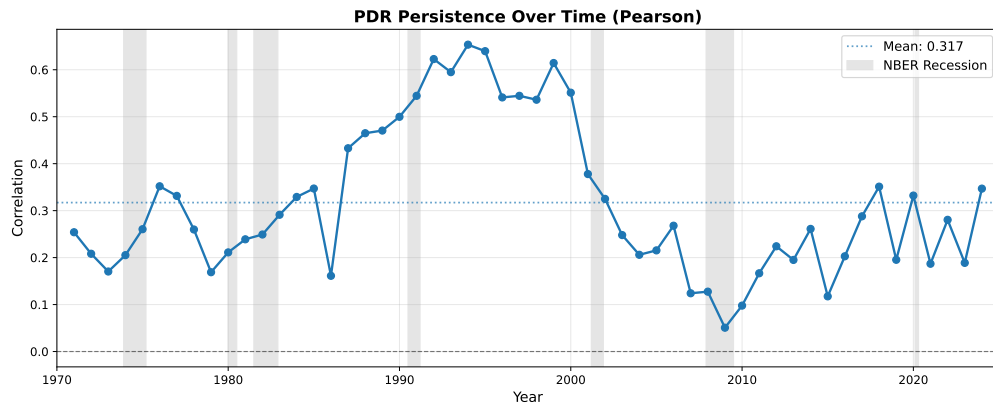


Table 1: PDR Persistence Comparison

This table compares the persistence of the Pattern Distance Ratio (PDR) with that in a Random Walk placebo test. In the 1-Year, 2-Year, 3-Year, and 5-Year specifications, we split the sample into non-overlapping one-, two-, three-, and five-year blocks, respectively. This gives 54, 27, 18, and 10 consecutive block-pairs. The 5-Year sample is smaller because its final block, 2021–2025, has no following block. The other specifications instead end in a one-year 2025 block. For each block, we correlate the in-sample PDR of its segments with the same segments’ out-of-sample PDR, which is measured against the next block. Note that a longer block changes *two* things at once: the temporal distance between the compared blocks and the size of the comparison sample from which nearest neighbors are drawn. Panel A reports the mean PDR by in-sample quintile under each specification (Q1 = most recurrent, Q5 = most novel). Panel B reports the overall (pooled) correlations between in-sample and out-of-sample PDR. Panel C reports the mean of the correlations computed separately for each block-pair. Panel D isolates the effect of temporal distance alone. It holds the comparison block fixed at one year and varies only the horizon h . That is, it correlates the in-sample PDR of annual block $\mathcal{D}_y$ with the out-of-sample PDR measured against $\mathcal{D}_{y+h}$ for $h = 1, 2, 3, 5$. The Random Walk column is a single realization of a shuffled benchmark. Because its persistence is itself a random variable, the informative comparison with the real-data columns is one of magnitude (Section 3.1). All analyses use data from 1971–2025.

Panel A: Mean PDR by In-Sample Quintile					
Quintile	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Q1	0.982	0.984	0.984	0.984	0.990
Q2	1.000	0.999	0.999	0.998	0.997
Q3	1.009	1.007	1.006	1.005	1.000
Q4	1.019	1.016	1.014	1.012	1.003
Q5	1.038	1.036	1.035	1.034	1.010
All segments	1.010	1.008	1.008	1.007	1.000
Panel B: Overall Correlation—Pooled					
Metric	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Pooled Pearson r	0.308	0.434	0.486	0.603	0.011
Pooled Spearman r	0.251	0.323	0.356	0.444	0.008
N samples	1,295,590	1,295,590	1,295,590	1,200,545	1,295,608
Panel C: Block-Pair Consistency					
Metric	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Mean Spearman	0.233	0.323	0.361	0.450	0.010
Panel D: Fixed Annual Blocks, Varying Horizon					
Horizon h	1 year	2 years	3 years	5 years	
Pooled Pearson r	0.308	0.275	0.270	0.257	
Pooled Spearman r	0.251	0.220	0.216	0.193	
Mean Spearman	0.233	0.218	0.207	0.198	
N samples	1,295,590	1,272,410	1,248,355	1,200,545	

Table 2: PDR Quintile Transition Matrices

This table presents transition matrices. Each cell shows the probability that a segment in quintile q_1 (in-sample) moves to quintile q_2 (out-of-sample). Rows are in-sample quintiles, and columns are out-of-sample quintiles. Higher diagonal values indicate stronger persistence. Panels A–D use one-, two-, three-, and five-year comparison blocks, and Panel E shows the Random Walk placebo. All analyses use data from 1971–2025.

Panel A: 1-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.269	0.234	0.196	0.168	0.133
Q2	0.229	0.248	0.227	0.183	0.112
Q3	0.196	0.222	0.229	0.212	0.140
Q4	0.172	0.181	0.208	0.232	0.206
Q5	0.134	0.113	0.139	0.204	0.408
Panel B: 2-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.296	0.248	0.197	0.157	0.102
Q2	0.233	0.257	0.230	0.182	0.099
Q3	0.191	0.223	0.235	0.218	0.134
Q4	0.162	0.174	0.211	0.245	0.207
Q5	0.118	0.099	0.127	0.199	0.458
Panel C: 3-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.310	0.251	0.193	0.152	0.094
Q2	0.238	0.259	0.232	0.179	0.092
Q3	0.189	0.227	0.240	0.219	0.125
Q4	0.155	0.173	0.214	0.254	0.204
Q5	0.108	0.091	0.121	0.196	0.485
Panel D: 5-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.345	0.256	0.192	0.142	0.066
Q2	0.253	0.262	0.230	0.177	0.078
Q3	0.192	0.231	0.241	0.222	0.114
Q4	0.142	0.175	0.223	0.264	0.196
Q5	0.068	0.076	0.114	0.196	0.546
Panel E: Random Walk					
	Q1	Q2	Q3	Q4	Q5
Q1	0.226	0.186	0.183	0.185	0.220
Q2	0.187	0.209	0.210	0.208	0.186
Q3	0.183	0.211	0.213	0.209	0.183
Q4	0.186	0.207	0.211	0.210	0.186
Q5	0.217	0.186	0.183	0.188	0.226

Table 3: Sign Preservation Rate by PDR Quintile and Neighbor-Outcome Score

This table presents sign preservation rates across Pattern Distance Ratio (PDR) quintiles for different block lengths. Panel A reports results for segments with high neighbor-outcome scores ($|T| \geq 1.96$), and Panel B for segments with low scores ($|T| < 1.96$). Appendix B explains how to interpret the cutoff. Sign preservation is the share of segments whose T-statistic has the same sign in-sample and out-of-sample. Q1 contains segments with the lowest PDR (highest recurrence), and Q5 contains segments with the highest PDR (lowest recurrence). The “Adj. Null” row reports the independence-adjusted null benchmark. It is computed as $P(\text{in}>0) \times P(\text{out}>0) + P(\text{in}<0) \times P(\text{out}<0)$, where $P(\text{out}>0)$ is the unconditional probability from the full sample. The blocks are one, two, three, and five years long, as in Table 1. The sample period is 1971–2025.

Panel A: High Neighbor-Outcome Score ($T \geq 1.96$)				
Quintile	1-Year	2-Year	3-Year	5-Year
Q1 (Low PDR)	53.07%	53.44%	52.50%	52.15%
Q2	53.56%	54.13%	53.21%	53.34%
Q3	53.08%	53.19%	52.73%	52.47%
Q4	52.84%	51.79%	52.40%	51.37%
Q5 (High PDR)	50.63%	51.21%	50.79%	50.51%
Adj. Null	50.24%	50.29%	50.39%	50.38%
Panel B: Low Neighbor-Outcome Score ($T < 1.96$)				
Quintile	1-Year	2-Year	3-Year	5-Year
Q1 (Low PDR)	51.13%	51.16%	51.00%	51.06%
Q2	51.11%	51.26%	51.13%	50.74%
Q3	51.04%	50.86%	50.77%	50.53%
Q4	50.92%	50.56%	50.56%	50.49%
Q5 (High PDR)	50.46%	49.83%	50.07%	50.23%
Adj. Null	50.02%	50.02%	50.02%	50.02%

Table 4: Long–Short Alphas: Factor Models and Subsample Stability

This table reports annualized alphas (%) of the long–short portfolio ($Q5 - Q1$) of Section 5.1. Portfolios are equal-weighted or value-weighted. Each panel reports alphas from seven models: the CAPM, the Fama–French three- and five-factor models (FF3, FF5) (Fama and French, 1993, 2015), the Carhart four-factor model (FF4) (Carhart, 1997), FF5 plus momentum (FF6), FF6 plus French’s monthly short-term reversal factor (ST-Rev), and that model plus a weekly reversal factor (WST-Rev). We build WST-Rev in our universe. Each month, we sort segments on the compound return over the last five days of the formation window, buy the bottom quintile, sell the top quintile, and hold for twenty days. The factor is equal-weighted in the equal-weighted column and value-weighted in the value-weighted column. Panel A covers the out-of-sample period 1975–2025, and Panels B and C cover its 2000–2025 and 2009–2025 subsamples. t -statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Factor models, full sample (1975–2025)		
Model	Equal-weighted	Value-weighted
CAPM	9.57*** (10.96)	6.84*** (5.76)
FF3	9.44*** (10.80)	6.69*** (5.63)
FF4	8.86*** (10.07)	6.43*** (5.32)
FF5	8.79*** (9.86)	6.07*** (4.99)
FF6	8.39*** (9.38)	5.92*** (4.81)
FF6 + ST-Rev	8.54*** (9.52)	5.81*** (4.70)
FF6 + ST-Rev + WST-Rev	7.30*** (7.69)	6.33*** (4.96)
Panel B: Factor models, 2000–2025		
Model	Equal-weighted	Value-weighted
CAPM	10.25*** (7.01)	9.17*** (4.78)
FF3	10.02*** (6.90)	8.87*** (4.67)
FF4	9.63*** (6.65)	8.77*** (4.59)
FF5	9.00*** (6.09)	8.05*** (4.13)
FF6	8.78*** (5.96)	8.03*** (4.09)
FF6 + ST-Rev	8.81*** (5.99)	8.00*** (4.09)
FF6 + ST-Rev + WST-Rev	8.19*** (5.50)	8.51*** (4.33)
Panel C: Factor models, 2009–2025		
Model	Equal-weighted	Value-weighted
CAPM	9.00*** (5.34)	8.74*** (4.35)
FF3	8.98*** (5.27)	8.65*** (4.27)
FF4	8.73*** (5.19)	8.32*** (4.17)
FF5	8.66*** (5.05)	8.42*** (4.11)
FF6	8.45*** (4.99)	8.12*** (4.04)
FF6 + ST-Rev	8.29*** (4.88)	7.87*** (3.92)
FF6 + ST-Rev + WST-Rev	8.43*** (4.96)	8.29*** (4.17)

Table 5: Signal Quality by Firm Size

This table reports the accuracy of the signal within firm-size terciles. The tercile breakpoints are the 33rd and 67th percentiles of NYSE market equity each month, applied to all stocks. The rank information coefficient (*rank-IC*) is the monthly unweighted Spearman correlation between the signal and the subsequent raw 20-day return across all stocks in the tercile. *Sharpe* is the annualized Sharpe ratio of the long–short portfolio formed inside the tercile, equal-weighted (EW) and value-weighted (VW). Results cover the full out-of-sample period (1975–2025) and its 2000–2025 and 2009–2025 subsamples. Newey–West (3 lags) *t*-statistics of the mean monthly rank-IC in parentheses; the Sharpe ratios carry no *t*-statistic. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Small	Mid	Large
<i>Full (1975–2025)</i>			
rank-IC	0.028*** (12.6)	0.023*** (7.7)	0.021*** (5.1)
Sharpe (EW/VW)	1.51/1.32	0.99/0.96	0.67/0.57
<i>2000–2025</i>			
rank-IC	0.025*** (6.6)	0.025*** (5.3)	0.023*** (3.6)
Sharpe (EW/VW)	1.25/1.08	0.89/0.88	0.65/0.73
<i>2009–2025</i>			
rank-IC	0.022*** (4.7)	0.023*** (4.0)	0.028*** (3.6)
Sharpe (EW/VW)	1.07/0.83	0.99/0.98	0.70/0.89

Table 6: Is the Signal Spanned by Technical Rules and Standard Predictors?

This table reports monthly Fama–MacBeth regressions of the 20-day holding return (in percent) on the signal and competing predictors, in the layout of Han et al. (2016). *Signal* is the within-month percentile rank of the engine’s score. Column (1) adds two moving-average rules built from the same 60-day window (Brock et al., 1992; Han et al., 2013): *MA ratio* is the last price over the 60-day average, and *MA crossover* is the 5-day over the 50-day average. Column (2) adds the standard predictors (De Bondt and Thaler, 1985; Ang et al., 2006; Bali et al., 2011; George and Hwang, 2004) and five liquidity measures averaged over the prior twelve months; *Bid–ask spread* is the average of the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) high–low estimators. Column (3) adds the last-five-day return. Column (4) adds the trend forecast (Han et al., 2016) and the price delay (Hou and Moskowitz, 2005) from the open-source panel of Chen and Zimmermann (2022). Regressors are winsorized and standardized within the month, so a slope is the change in the 20-day return, in percent, per one-standard-deviation move. The regressions are equal-weighted, and Table A7 reports the value-weighted version. Columns (1)–(3) cover 1975–2025 (611 months, 1,673 stocks per month), and column (4) ends in 2024 (595 months, 1,610 stocks per month). Newey–West (12 lags) *t*-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal (engine prediction)	0.193*** (7.09)	0.152*** (8.21)	0.182*** (8.33)	0.155*** (7.09)
MA ratio	−1.076*** (−8.06)	−1.011*** (−8.62)	−0.408*** (−4.57)	−0.334*** (−3.58)
MA crossover	0.897*** (7.22)	0.895*** (8.66)	0.321*** (3.80)	0.220** (2.29)
Reversal (monthly, 21-day)		−0.275*** (−5.53)	−0.160*** (−3.42)	−0.108** (−2.24)
Momentum (12–2)		0.243*** (3.45)	0.232*** (3.35)	0.268*** (5.12)
Long-term reversal (60–25)		−0.042 (−1.38)	−0.042 (−1.42)	−0.046 (−1.59)
IVOL		−0.240*** (−4.46)	−0.242*** (−4.58)	−0.288*** (−5.35)
Max		0.026 (0.92)	0.034 (1.24)	0.065*** (2.81)
Beta		0.062 (0.95)	0.065 (1.06)	0.061 (0.91)
log market equity		−0.342*** (−3.34)	−0.361*** (−3.53)	−0.288*** (−3.32)
log book-to-market		0.026 (0.58)	0.021 (0.47)	0.014 (0.32)
Illiquidity (Amihud)		0.069*** (2.87)	0.073*** (3.01)	0.087*** (3.96)
Turnover		−0.052 (−0.96)	−0.060 (−1.15)	−0.059 (−1.14)
Bid–ask spread		0.049 (1.05)	0.036 (0.79)	0.024 (0.52)
Dollar volume (log)		0.213** (2.17)	0.233** (2.39)	0.178** (2.19)
Zero-volume days		−0.017 (−1.55)	−0.017 (−1.53)	−0.028** (−2.30)
Trend forecast (HZZ)				0.077* (1.75)
Price delay				0.001 (0.05)
52-week high		0.044 (0.88)	0.019 (0.40)	−0.052 (−1.15)
Reversal (weekly, 5-day)			−0.298*** (−6.72)	−0.258*** (−6.47)
Avg. adj. R^2 (%)	2.44	10.05	10.28	10.86

Table 7: Implementability: Turnover and Break-Even Trading Costs

This table reports the strategy's turnover and the trading cost that would use up its alpha. Panel A is the baseline monthly-rebalanced strategy of Table 4. Panel B restricts both legs to the largest NYSE size tercile of Table 5. EW and VW rows are equal- and value-weighted. *Gross* is the annualized FF6 alpha (%) with its *t*-statistic in parentheses. *Turnover* is one-way turnover per leg, in percent per month. The break-even columns report, in basis points, a uniform one-way cost per dollar traded. *Raw* is the cost at which the raw long–short return is zero. It equals the average monthly long–short return divided by the dollars traded per month per dollar held on a side, counting every purchase and sale on both legs. The $\alpha = 0$ column is the cost at which the net FF6 alpha is zero. It equals the gross FF6 alpha divided by the same dollars traded. The $t = 1.96$ column is the cost at which the *t*-statistic of the net alpha falls to 1.96. The round-trip cost is twice the figure shown. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Monthly rebalancing, all stocks

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	Break-even one-way cost (bp)		
				Raw	$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	8.39*** (9.38)	82	23.2	21.3	16.7
	VW	5.92*** (4.81)	84	15.9	14.6	8.6
2000–2025	EW	8.78*** (5.96)	82	24.0	22.3	15.0
	VW	8.03*** (4.09)	84	20.6	19.9	10.4
2009–2025	EW	8.45*** (4.99)	82	21.3	21.4	13.0
	VW	8.12*** (4.04)	84	19.5	20.2	10.4

Panel B: Monthly rebalancing, largest NYSE size tercile

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	Raw	$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	5.01*** (4.04)	82	14.3	12.7	6.5
	VW	4.43*** (3.40)	83	12.7	11.2	4.7
2000–2025	EW	5.97*** (2.99)	82	16.5	15.2	5.3
	VW	7.01*** (3.41)	82	18.9	17.7	7.5
2009–2025	EW	5.08** (2.38)	82	14.7	12.9	2.3
	VW	6.88*** (3.38)	82	18.1	17.4	7.3

Table 8: The Engine in International Markets: Out-of-Sample Deployment

This table reports results for the engine of Section 5.5 in the twelve international markets of Table A5. *Panel A* reports **annualized** long–short ($Q5 - Q1$) alphas (%), equal- and value-weighted, for the full screened universe. Each market is benchmarked against its own regional five-factor-plus-momentum model (Fama and French, 2017). Markets are grouped into French’s Developed and Emerging universes and ordered within each group by Stage I PDR persistence. *Global* is the country-average portfolio of Chui et al. (2010), reported as a raw return. *Developed* and *Emerging* split it by universe and are alphas against their own regional models. Panel A t -statistics are from ordinary least squares. *Panel B* reports pooled cross-country Fama–MacBeth regressions of the 20-day forward return, demeaned within each country-month, on within-country-month percentile ranks of the signal and the controls. The coefficients are per holding period, not annualized. When Compustat Global book equity is missing or negative (25% of firm-months), book-to-market is set to zero and an availability indicator (B/M available) is added (McLean et al., 2009). The last column adds Amihud (2002) illiquidity. Panel B t -statistics are Newey–West, from 396 monthly slopes (395 in the last column). ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Long–short returns by market

Market	Months	EW	VW
<i>Developed markets</i>			
United Kingdom	324	3.01** (2.05)	6.91** (2.53)
Hong Kong	276	10.36*** (5.30)	9.20*** (3.32)
Japan	396	5.79*** (5.73)	5.56*** (3.60)
France	312	4.30*** (2.79)	1.98 (0.63)
Australia	264	10.52*** (4.49)	3.14 (1.09)
Sweden	204	12.82*** (5.24)	7.75** (2.23)
<i>Emerging markets</i>			
Thailand	216	3.02 (1.50)	8.20** (2.21)
Malaysia	288	7.38*** (3.70)	11.56*** (4.49)
South Korea	300	2.81* (1.67)	0.06 (0.02)
India	252	11.36*** (5.91)	3.63 (1.16)
Taiwan	324	1.41 (0.67)	−0.62 (−0.25)
China	245	3.43* (1.71)	1.80 (0.53)
Global, raw return	300	6.19*** (11.13)	4.92*** (5.70)
Developed, regional-factor alpha	312	8.12*** (10.36)	6.91*** (5.34)
Emerging, regional-factor alpha	288	5.92*** (6.88)	4.51*** (3.55)

Panel B: Pooled cross-country Fama–MacBeth regressions

	Univariate	Multivariate	+ Weekly rev.	+ Illiquidity
Signal (engine prediction)	0.63*** (9.75)	0.57*** (10.00)	0.58*** (10.05)	0.59*** (10.10)
log market equity		−0.84*** (−3.86)	−0.80*** (−3.68)	−0.47 (−1.42)
Book-to-market		0.66*** (3.21)	0.66*** (3.25)	0.63*** (3.19)
B/M available		0.40*** (3.56)	0.39*** (3.57)	0.39*** (3.51)
Reversal (monthly)		−0.61*** (−3.05)	−0.19 (−1.10)	−0.17 (−0.95)
Reversal (weekly)			−0.99*** (−6.46)	−1.01*** (−7.03)
Momentum (12–2)		0.54* (1.87)	0.57** (2.01)	0.61** (2.16)
Illiquidity (Amihud)				0.45* (1.73)

Appendix

A. Summary of Variables and Notations

Table A1: Summary of Variables and Notations

Symbol	Description
Data and Sample Construction	
$r_{i,t}$	Daily return of stock i on day t
$P_{j,t}$	Closing price of stock j on day t
$SHROUT_{j,t}$	Shares outstanding of stock j on day t
$ME_{j,t}$	Market equity: $ P_{j,t} \times SHROUT_{j,t}$
BE	Book equity: $SEQ + TXDITC - PS$
BM	Book-to-market ratio: $BE_{y-1} / ME_{Dec,y-1}$
$p(j)$	FF25 portfolio to which stock j is assigned
Return Segment Construction and Standardization	
$\mathbf{R}_{i,t}$	60-day return segment for stock i ending at date t : ($r_{i,t-59}, \dots, r_{i,t}$)
$\bar{R}_n$	Within-window mean: $(1/60) \sum_{s=t-59}^t r_{i,s}$
σ_n	Within-window standard deviation
$\tilde{R}_{n,k}$	Standardized return: $(R_{n,k} - \bar{R}_n) / \sigma_n$
Distance and Nearest Neighbors	
$d_{m,n}$	Z-normalized Euclidean distance between standardized segments m and n (Equation 2)
K	Number of nearest neighbors ($K = 100$)
v_k^{real}	k -th nearest real-data neighbor of segment n
$v_k^{\text{rand},m}$	k -th nearest neighbor from random walk series m
$\mathcal{D}_y$	Set of all segments whose formation window ends in year y
Random Walk Benchmark	
$\mathcal{R}_y$	Pool of all daily returns across all stocks in year y
N_y	Total number of return observations in year y : $ \mathcal{R}_y $
Pattern Distance Ratio (PDR)	
$\bar{d}_n^{\text{real} \rightarrow \text{real}}$	Mean distance from segment n to its K nearest real neighbors
$\bar{d}_n^{\text{real} \rightarrow \text{rand}}$	Mean distance from segment n to its K nearest neighbors in each random walk series, averaged across the 3 series
PDR_n	Pattern Distance Ratio: $\bar{d}_n^{\text{real} \rightarrow \text{real}} / \bar{d}_n^{\text{real} \rightarrow \text{rand}}$
Return Measures and Predictability	
R_n^{FF25}	FF25-adjusted cumulative 20-day return for segment n

Continued on next page

Table A1 – continued from previous page

Symbol	Description
R_n^{raw}	Unadjusted cumulative 20-day return
$r_{p(j),t+s}$	Daily return of the FF25 benchmark portfolio for stock j on day $t + s$
T_n	T-statistic of the mean future FF25-adjusted return of segment n 's K nearest neighbors
$\bar{R}_n^{\text{FF25}}$	Mean future FF25-adjusted return of segment n 's K nearest neighbors
$\hat{\sigma}_n^{\text{FF25}}$	Standard deviation of the future FF25-adjusted returns of segment n 's K nearest neighbors
Persistence Testing	
PDR_n^{in}	In-sample PDR (from $\mathcal{D}_y$ vs. $\mathcal{D}_y$)
$PDR_n^{\text{out},h}$	Out-of-sample PDR at horizon h (from $\mathcal{D}_y$ vs. $\mathcal{D}_{y+h}$)
$T_n^{\text{in}}, T_n^{\text{out},h}$	In-sample and out-of-sample T-statistics
$\mathbf{T}^h$	PDR quintile transition matrix for block length h
T_{q_1, q_2}^h	Transition probability from quintile q_1 to q_2
SP^h	Sign preservation rate for block length h
PDR_n^{placebo}	Placebo PDR: $\bar{d}_n^{\text{rand}, 4 \rightarrow 4} / \bar{d}_n^{\text{rand}, 4 \rightarrow \text{rand}}$
Portfolio Construction (Appendix B)	
S_Y	Landmark set for test year Y (the recurrence grid). The pool holds the segments from years $Y - 3$ to $Y - 1$ with $ T \geq 1.96$. We draw 410 segments from each PDR decile of this pool
$R_{q,\tau}^{\text{EW}}, R_{q,\tau}^{\text{VW}}$	Equal-weighted and value-weighted portfolio returns for quintile q
$R_{LS,\tau}$	Long–short portfolio return (Q5 – Q1)
Factor Models	
$MktRf_\tau$	Market excess return
SMB_τ, HML_τ	Size and value factors
UMD_τ	Momentum factor
RMW_τ, CMA_τ	Profitability and investment factors

B. Portfolio Construction Methodology

This appendix provides the formal definitions of the return measures, the block-bootstrap surrogate benchmark of Section 3.3, the segment-selection filters that define the recurrence grid, and the portfolio-formation and factor-model conventions used in Section 5.

Return Measures

For each return segment n associated with stock j , we compute the Fama–French 25 size-BM characteristic-adjusted return (Fama and French, 1993) over the 20-day *evaluation window* immediately following the *formation window*:

$$R_n^{FF25} = \prod_{s=1}^{20} (1 + r_{j,t+s}) - \prod_{s=1}^{20} (1 + r_{p(j),t+s}), \quad (8)$$

where t is the last day of the *formation window* and $r_{p(j),t+s}$ is the daily value-weighted return of the FF25 benchmark portfolio to which stock j is assigned. We evaluate predictability by examining the future FF25-adjusted returns of each segment’s $K = 100$ nearest neighbors. The T-statistic is:

$$T_n = \frac{\bar{R}_n^{FF25}}{\hat{\sigma}_n^{FF25}/\sqrt{K}}, \quad (9)$$

where $\bar{R}_n^{FF25}$ and $\hat{\sigma}_n^{FF25}$ are the sample mean and standard deviation of the neighbors’ future FF25-adjusted returns. A large $|T_n|$ means that segments similar to n have been followed by abnormal returns in a consistent direction. These FF25-adjusted returns are the main outcome variable. We use them to compute the T-statistics above, and, after the reversal residualization and rank transform of Section 5, they are also the prediction target for the engine.

An important caveat applies to how we interpret T_n throughout the paper. The neighbors’ future returns are not independent observations. Some neighbors share window end dates, and in that case their 20-day evaluation windows overlap entirely. Neighbors also load on common shocks beyond the FF25 adjustment, and they occasionally include the same stock at nearby dates. The $\sqrt{K}$ scaling therefore overstates precision, and the cutoff $|T_n| \geq 1.96$ does not carry a nominal five-percent size. For this reason, we treat T_n as a signal-to-noise *score* that ranks segments. We use 1.96 only as a fixed, conventional cutoff chosen ex ante, and no result in the paper depends on its calibration.

The Block-Bootstrap Surrogate Benchmark

The surrogate benchmark in Table A2 replaces the cross-stock shuffle with a different null. This null keeps each stock’s own dynamics and the market’s factor structure, and it randomizes only the idiosyncratic shape. We build it in four steps.

Factor exposures. For each stock i and each calendar year y we regress the stock’s daily excess return on the six Fama–French factors (FF5 plus UMD) and a constant, using that year’s daily observations: $r_{i,t} - r_{f,t} = a_{i,y} + \hat{b}'_{i,y} F_t + e_{i,t}$. We estimate exposures year by year rather than over the full sample. This lets a stock’s factor loadings drift over time and absorbs each year’s comovement locally. We require more than 200 trading days in the stock-year. Stock-years below that threshold have no estimated beta, so they contribute no surrogate values. As we

explain below, the cost of this rule falls on the benchmark pool rather than on the segments being measured.

Idiosyncratic series. The idiosyncratic return is $e_{i,t} = r_{i,t} - \hat{b}'_{i,y} F_t$, the raw return net of the fitted factor component only. We leave in the intercept and the risk-free rate, because every 60-day segment is z-normalized before any distance is computed. That step removes any constant level.

Resampling. We resample each stock's $e_{i,t}$ independently, using the stationary bootstrap of Politis and Romano (1994). Block lengths are geometric with success probability $p = 1/L$, so the mean block length is L trading days. Resampling is circular. We draw blocks within each maximal run of consecutive trading days for which the stock has a valid beta-year. Such a run typically spans several years. We then write the resampled values back to the same calendar dates. Scrambling each stock independently is harmless here, because the factor term carries the entire common component, and we do not resample that term.

Reassembly. The surrogate return is $\tilde{r}_{i,t} = \hat{b}'_{i,y} F_t + \tilde{e}_{i,t}$. We add the factor component back, and we keep the realized factor series F_t in its original time order, attached to its own date. Every surrogate stock therefore carries the same real factor path as the data. By construction, the level and timing of comovement in the benchmark panel are identical to those in the real panel. Only the idiosyncratic shape sequence is randomized. We generate three independent replicates, which take the place of the three random-walk benchmark panels. This count is inherited from the baseline design rather than chosen here. The shuffle benchmark averages over exactly three random-walk panels. A denominator averaged over more draws would carry less sampling noise, so the ratios would differ for a reason unrelated to the null, and the null is precisely what Panel A is built to compare. The ratio therefore keeps the form of Equation (6), $PDR_n = 3 \bar{d}_n^{\text{real} \rightarrow \text{real}} / \sum_{k=1}^3 \bar{d}_n^{\text{real} \rightarrow \text{sur},k}$, where each denominator term is the mean distance to the $K = 100$ nearest neighbors within replicate k . Each replicate supplies one surrogate path for every stock-day that has an estimated beta. As a result, the benchmark pool is the same panel of stocks and dates as the real cross-section, minus the stock-years that the 200-day rule removes. It holds 11.0% fewer complete 60-day segments (1.15 against 1.30 million). The real segments are unaffected, because each one is still a query against the pool. Of the real segments, 99.6% enter the surrogate analyses.

One difference in granularity is worth making explicit. For each replicate, the bootstrap runs once over the whole panel, not year by year, and it draws blocks along a stock's own history. The shuffle benchmark it replaces instead permutes returns across stocks within a single year. The two nulls are aligned where the comparison happens, not where the randomization happens. The surrogate panel is cut into the same annual blocks $\mathcal{D}_y$, carries the same segment identifiers as the real panel, and runs through the same distance and neighbor code. The PDR values are therefore directly comparable. The main specification sets $L = 10$ trading days, and we report $L = 20$ alongside it.

Landmark Selection (the Recurrence Grid)

For each test year Y , we draw the landmark grid from the pool of segments in the trailing three years ($Y - 3$ through $Y - 1$). The selection uses point-in-time statistics, measured against those years' own comparison blocks:

1. **Score filter** ($|T_n| \geq 1.96$). We keep segments whose neighbor T-statistic (Equation 9) satisfies $|T_n| \geq 1.96$. As a result, measurement points are placed only where the local region of trajectory space has shown a high neighbor-outcome score.

2. **Recurrence stratification (410 per PDR decile).** We split the retained pool at its own PDR deciles and draw 410 segments at random from each decile. The grid thus samples the recurrent and the novel regions of trajectory space evenly, rather than concentrating on either one. The draw gives up to 4,100 landmarks per test year. We obtain 4,100 in forty-nine of the fifty-one test years, and 1,612 and 3,239 in the two earliest years, where a decile holds fewer than 410 segments. We would skip any year with fewer than 1,000 qualifying segments, but no year is skipped.

Because the draw is random within deciles, we draw five independent grids per test year. We then average the five resulting predictions by within-month rank, so that no single draw determines the signal. We denote a resulting landmark set by S_Y and call it the *recurrence grid*.

Portfolio Formation

On each month-end date in the test year, we sort all segment observations by the model’s predicted score and assign them to five quintile portfolios, Q1 (lowest) through Q5 (highest). We construct two types of portfolios:

- **Equal-weighted (EW):** The portfolio return for quintile q on month-end date τ is the simple average of the holding-period returns of all segments assigned to quintile q :

$$R_{q,\tau}^{EW} = \frac{1}{|Q_{q,\tau}|} \sum_{n \in Q_{q,\tau}} R_n^{raw}, \quad (10)$$

where $Q_{q,\tau}$ denotes the set of segments in quintile q on date τ , and $R_n^{raw} = \prod_{s=1}^{20} (1 + r_{j,t+s}) - 1$ is the unadjusted cumulative holding-period return. We use *raw* (unadjusted) returns to evaluate portfolios because these are the actual returns an investor would earn. We use the FF25-adjusted returns only to build the prediction target (Section 5), so that the model targets abnormal performance rather than systematic factor exposures.

- **Value-weighted (VW):** The portfolio return for quintile q on date τ is the weighted average of holding-period returns, where the weights are market capitalizations:

$$R_{q,\tau}^{VW} = \frac{\sum_{n \in Q_{q,\tau}} w_n \cdot R_n^{raw}}{\sum_{n \in Q_{q,\tau}} w_n}, \quad (11)$$

where w_n is the market equity of segment n ’s stock at the beginning of the *evaluation window*. To mitigate the influence of extreme weights, we winsorize market capitalizations at the 1st and 99th percentiles within each date before computing the weighted average.

The **long–short portfolio** is Q5 minus Q1. It goes long the highest-scored quintile and short the lowest.

Factor Model Specifications

We evaluate the long–short portfolio’s risk-adjusted performance by regressing its monthly returns on standard factor models. Let $R_{LS,\tau}$ denote the long–short portfolio return for month τ . We estimate alphas from the following seven models:

1. **CAPM:** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_{\tau} + \varepsilon_{\tau}$
2. **Fama–French 3-factor (FF3):** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_{\tau} + \beta_{smb} \cdot SMB_{\tau} + \beta_{hml} \cdot HML_{\tau} + \varepsilon_{\tau}$
3. **Carhart 4-factor (FF4):** $FF3 + \beta_{umd} \cdot UMD_{\tau}$
4. **Fama–French 5-factor (FF5):** $FF3 + \beta_{rmw} \cdot RMW_{\tau} + \beta_{cma} \cdot CMA_{\tau}$
5. **6-factor (FF6):** $FF5 + \beta_{umd} \cdot UMD_{\tau}$
6. **FF6 + ST-Rev:** $FF6 + \beta_{strev} \cdot STRev_{\tau}$, where $STRev$ is French’s monthly short-term reversal factor
7. **FF6 + ST-Rev + WST-Rev:** $FF6 + STRev + \beta_{wstrev} \cdot WSTRev_{\tau}$, where $WSTRev$ is the weekly reversal factor built inside our universe under the strategy’s own portfolio rule (Table 4)

where $MktRf$ is the market excess return, SMB is the size factor, HML is the value factor, UMD is the momentum factor, RMW is the profitability factor, and CMA is the investment factor. All factor returns except WST-Rev come from Kenneth French’s data library. We construct WST-Rev as described in item 7 above. The intercept α in each regression is the abnormal return that the corresponding set of risk factors cannot explain. All returns and alphas are annualized by multiplying monthly values by 12, and Sharpe ratios are annualized by multiplying the monthly Sharpe ratio by $\sqrt{12}$.

C. Robustness

This appendix collects the robustness evidence for Stage I. The Stage II robustness checks are reported in Section 5.2. Tables A2 through A4 concern the recurrence measure itself. The recurrence measure rests on two choices: the random-walk benchmark and the nearest-neighbor rule. Table A2 replaces the benchmark with a stationary block bootstrap of factor-model residuals and replaces the neighbor rule with a diversified neighbor search. It then re-runs the downstream transition analysis against this harder null. Table A3 asks whether PDR persistence is a restatement of volatility persistence. Table A4 varies the length of the pattern window. Table A5 describes how the twelve international samples are built and reports the persistence measured in each. Table A6 varies the pattern window and the holding period together in the sign-preservation test. Finally, one Stage II table also appears here: Table A7 reports the value-weighted version of the horse race in Table 6.

Table A2: Robustness: PDR Persistence and Transition Matrix under Alternative Benchmark and Neighbor Definitions

This table checks PDR persistence under two alternative constructions of the PDR. Panel A reports, for each construction, the mean Pearson and Spearman correlations between in-sample and out-of-sample PDR across annual year-pairs at the 1-year horizon. *Strict neighbors* selects the $K = 100$ nearest neighbors with at most one neighbor per size $\times$ book-to-market $\times$ 48-industry $\times$ month cell. *Block-bootstrap surrogate* replaces the cross-stock-shuffle benchmark with a stationary block bootstrap of each stock's six-factor (FF5+UMD) residuals, and L is its mean block length in trading days. Appendix B gives the full construction. The Baseline and Random walk placebo rows repeat the main results. Panel B reports the 1-year quintile transition matrix of the PDR measured against the $L = 10$ surrogate. Its counterpart under the shuffle benchmark is Panel A of Table 2. All results use the 60-day pattern window and the 1971–2025 sample.

Panel A: PDR persistence under alternative constructions					
Construction		Mean Pearson r		Mean Spearman r	
Baseline		0.317		0.233	
Strict neighbors		0.350		0.250	
Block-bootstrap surrogate ($L = 10$)		0.247		0.176	
Block-bootstrap surrogate ($L = 20$)		0.262		0.184	
Random walk placebo		0.014		0.010	
Panel B: 1-year transition matrix, surrogate-based PDR ($L = 10$)					
	Q1	Q2	Q3	Q4	Q5
Q1	0.362	0.195	0.160	0.143	0.140
Q2	0.194	0.226	0.217	0.200	0.163
Q3	0.158	0.215	0.224	0.219	0.185
Q4	0.143	0.200	0.216	0.225	0.216
Q5	0.143	0.165	0.184	0.213	0.296

Table A3: PDR Persistence and the Volatility Channel

The dependent variable is the one-year-ahead out-of-sample PDR, $PDR_n^{\text{out},1}$. The regressors are the segment's in-sample PDR and four controls measured on the same 60-day formation window. *IVOL* is the standard deviation of the residual from regressing the stock's daily return on the market return over those 60 days. *Log ME* is log market equity on the window's last day, *Reversal* is the compound return over its last 21 days, and *Trend* is the slope of the cumulative log-price path. Coefficients are Fama–MacBeth averages of the 54 year-pair cross-sectional regressions, with every variable standardized within each cross-section. Newey–West (3 lags) *t*-statistics in parentheses. Avg. R^2 is the mean of the per-cross-section R^2 . All three columns use one common sample. The lower panel reports the mean year-pair persistence correlation within in-sample IVOL terciles, formed separately in each year-pair. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	Dependent variable: $PDR_n^{\text{out},1}$		
	(1)	(2)	(3)
In-sample PDR (PDR_n^{in})	0.317*** (8.08)	0.304*** (7.87)	0.304*** (7.85)
IVOL		0.093*** (9.11)	0.095*** (7.28)
Log ME			0.003 (0.26)
Reversal (monthly, 21-day)			0.017*** (3.26)
Trend			0.002 (0.20)
Avg. R^2	0.125	0.136	0.144
Observations	1,295,590	1,295,590	1,295,590
<i>Mean persistence correlation within IVOL terciles</i>			
Low IVOL	0.215	(positive in 49 of 54 year-pairs)	
Mid IVOL	0.270	(positive in 53 of 54 year-pairs)	
High IVOL	0.389	(positive in 54 of 54 year-pairs)	

Table A4: Robustness: PDR Persistence by Pattern Window Length

This table reports PDR persistence over the grid of pattern windows {30, 60, 90} days and holding periods {20, 30} days (columns 20d and 30d). The baseline is the 60-day window with a 20-day holding period. The holding period does not enter the PDR. It changes only which segments qualify, because a segment needs complete data for the pattern window plus the holding period. Panel A reports the pooled Pearson and Spearman correlations between in-sample and out-of-sample PDR, with N in thousands of segments. Panel B reports year-by-year consistency: the mean and standard deviation of the yearly Spearman correlations. All specifications use one-year comparison blocks with data from 1971–2025.

Panel A: Overall Correlation—Pooled						
Metric	30-day pattern		60-day pattern		90-day pattern	
	20d	30d	20d	30d	20d	30d
Pearson r	0.333	0.330	0.308	0.304	0.287	0.287
Spearman r	0.289	0.288	0.251	0.250	0.240	0.240
N (000s)	1,314	1,302	1,296	1,283	1,250	1,250
Panel B: Year-by-Year Consistency						
Metric	30-day pattern		60-day pattern		90-day pattern	
	20d	30d	20d	30d	20d	30d
Mean Spearman	0.274	0.273	0.233	0.232	0.209	0.209
Std Spearman	0.157	0.157	0.124	0.123	0.118	0.118

Table A5: International Samples and PDR Persistence Across Markets

This table describes the twelve international samples and reports PDR persistence in each market. The data and screens are described in Section 3.3. *Segments* is the number of 60-trading-day return windows anchored at month-ends with complete data. *Engine OOS* is the out-of-sample prediction window of the engine in Section 5.5, and *Stocks/month* is the average number of scored stocks per month within it. The PDR is built exactly as in the U.S. analysis, with the three-shuffle random-walk benchmark (Section 2.3). Persistence is the cross-sectional correlation between in-sample PDR and one-year-ahead out-of-sample PDR in each country-year. We retain country-years with at least 5,000 segments that have both an in-sample and an out-of-sample PDR. *Years* counts them, and *N* is their pooled number of segments. *Mean r* and *t* are the time-series mean and cross-year *t*-statistic of the annual Pearson correlations. *% Pos* is the share of country-years with a positive correlation. *Pooled r* and *Pooled ρ* are computed over the pooled cross-section of all retained country-years. Markets with fewer than five qualifying years are omitted: Indonesia has four, Turkey three, and eleven further candidate markets, including Germany, have none. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Market	Sample construction				PDR persistence					
	Segments	Stocks/ month	Engine OOS	Years	<i>N</i>	Mean <i>r</i>	<i>t</i>	% Pos	Pooled <i>r</i>	Pooled ρ
United Kingdom	249,102	692	1999–2025	30	240,496	0.488***	27.07	100.0	0.484	0.392
Hong Kong	209,142	669	2003–2025	15	155,737	0.481***	10.81	100.0	0.530	0.471
Thailand	124,276	462	2008–2025	11	74,818	0.430***	10.79	100.0	0.433	0.375
Malaysia	122,738	349	2002–2025	7	40,208	0.370***	4.58	100.0	0.362	0.320
Japan	967,615	2,341	1993–2025	36	952,840	0.361***	11.05	91.7	0.460	0.330
France	156,285	424	2000–2025	19	102,237	0.321***	14.09	100.0	0.311	0.255
South Korea	492,962	1,529	2001–2025	27	470,676	0.277***	8.08	92.6	0.405	0.318
Australia	128,415	400	2004–2025	7	41,948	0.261***	5.62	100.0	0.249	0.177
Sweden	93,338	340	2009–2025	6	34,934	0.191***	4.67	100.0	0.176	0.147
India	386,948	1,402	2005–2025	21	353,356	0.176***	7.12	95.2	0.198	0.186
Taiwan	401,613	1,151	1999–2025	24	359,546	0.146***	6.00	87.5	0.146	0.145
China	586,052	2,225	2005–2025	20	544,023	0.086***	2.98	70.0	0.141	0.126
Total	3,918,486	—	—	—	—	—	—	—	—	—

Table A6: Robustness: Sign Preservation Rate by Pattern Length and Holding Period

This table reports sign preservation rates by PDR quintile for the full grid of pattern windows {30, 60, 90} days and holding periods {20, 30} days (columns 20d and 30d). The baseline is the 60-day pattern with a 20-day holding period. Panel A covers segments with high neighbor-outcome scores ($|T| \geq 1.96$), and Panel B those with low scores ($|T| < 1.96$). Sign preservation is the share of segments whose T-statistic has the same sign in-sample and out-of-sample. Entries are in percent. The “Adj. Null” row is the independence-adjusted null benchmark, computed as in Table 3. All specifications use one-year blocks with data from 1971–2025.

Panel A: High Neighbor-Outcome Score ($T \geq 1.96$)						
Quintile	30-day pattern		60-day pattern		90-day pattern	
	20d	30d	20d	30d	20d	30d
Q1 (Low PDR)	53.40	52.09	53.07	52.44	52.22	51.12
Q2	53.42	51.96	53.56	53.00	52.19	51.30
Q3	52.98	51.74	53.08	52.42	52.06	51.49
Q4	52.85	51.83	52.84	51.87	51.57	50.93
Q5 (High PDR)	49.56	49.66	50.63	49.72	50.84	50.66
Adj. Null	50.22	50.18	50.24	50.17	50.11	50.26
Panel B: Low Neighbor-Outcome Score ($T < 1.96$)						
Quintile	30-day pattern		60-day pattern		90-day pattern	
	20d	30d	20d	30d	20d	30d
Q1 (Low PDR)	50.97	50.66	51.13	50.83	50.67	50.58
Q2	51.43	50.73	51.11	50.54	50.83	50.61
Q3	51.13	50.90	51.04	50.75	50.83	50.52
Q4	50.75	50.55	50.92	50.68	50.62	50.30
Q5 (High PDR)	50.38	50.21	50.46	50.31	50.48	50.21
Adj. Null	50.02	50.01	50.02	50.00	50.00	50.02

Table A7: The Horse Race, Value-Weighted

This table repeats Table 6 with value-weighted Fama–MacBeth regressions. We estimate each month’s cross-sectional regression by weighted least squares, using market equity as the weight (Medhat and Schmeling, 2022; Green et al., 2017). The columns, the sample, the variable definitions, the winsorization and standardization, and the Newey–West (12 lags) t -statistics are exactly the same as in Table 6. Only the weighting differs. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal (engine prediction)	0.139*** (3.54)	0.076*** (3.11)	0.089*** (3.55)	0.072*** (3.00)
MA ratio	−0.999*** (−5.29)	−0.974*** (−6.48)	−0.513*** (−3.29)	−0.495*** (−3.19)
MA crossover	0.936*** (4.94)	0.813*** (5.78)	0.376*** (2.70)	0.342** (2.32)
Reversal (monthly, 21-day)		−0.082 (−1.43)	−0.001 (−0.02)	0.013 (0.22)
Momentum (12–2)		0.373*** (4.83)	0.360*** (4.75)	0.387*** (5.40)
Long-term reversal (60–25)		−0.077* (−1.73)	−0.077* (−1.76)	−0.076* (−1.77)
IVOL		−0.284*** (−3.62)	−0.276*** (−3.62)	−0.289*** (−4.04)
Max		0.062 (1.43)	0.063 (1.47)	0.065 (1.61)
Beta		0.069 (0.79)	0.084 (1.00)	0.043 (0.51)
log market equity		0.068 (0.46)	0.058 (0.39)	0.030 (0.21)
log book-to-market		0.007 (0.15)	0.003 (0.07)	−0.009 (−0.21)
Illiquidity (Amihud)		0.162** (2.57)	0.160** (2.57)	0.161*** (2.59)
Turnover		0.123* (1.65)	0.122* (1.67)	0.123 (1.62)
Bid–ask spread		0.010 (0.16)	−0.013 (−0.20)	0.008 (0.15)
Dollar volume (log)		−0.119 (−0.88)	−0.113 (−0.85)	−0.088 (−0.65)
Zero-volume days		−0.045 (−1.47)	−0.044 (−1.49)	−0.047 (−1.60)
Trend forecast (HZZ)				0.012 (0.18)
Price delay				−0.004 (−0.13)
52-week high		−0.192** (−2.55)	−0.212*** (−2.82)	−0.220*** (−2.99)
Reversal (weekly, 5-day)			−0.216*** (−4.66)	−0.213*** (−4.23)
Avg. adj. R^2 (%)	5.57	20.20	20.65	21.41

D. A Latent-Template Model of Pattern Recurrence

This appendix formalizes the density-ratio interpretation of the PDR (Section 2.4) in a simple statistical model of recurring return trajectories, and it derives the empirical implications tested in Sections 3–5. The model is deliberately reduced-form. “Templates” stand for any economic mechanism that makes particular return dynamics repeat across stocks and over time, such as earnings and announcement cycles, institutional rebalancing, the gradual diffusion of information, or the crowding of trading strategies. The framework’s role is to state precisely *what* the PDR identifies, not to explain *why* recurrence arises.

Setup

After z-normalization, each m -day return segment ($m = 60$) is a point in the shape space

$$\mathcal{X} = \left\{ x \in \mathbb{R}^m : \sum_{i=1}^m x_i = 0, \|x\|^2 = m - 1 \right\},$$

a sphere of dimension $p = m - 2 = 58$. We model the segments in a comparison block as independent draws from a two-part mixture.

Assumption A1 (latent-template mixture). With probability $1 - \pi$, a segment is drawn from a diffuse null density f_0 on $\mathcal{X}$, the shape distribution produced by returns that have no time structure and are pooled across stocks. With probability π , it is a noisy realization of one of M latent templates: template k is chosen with weight w_k and generates the segment from a density f_k concentrated near a location $\tau_k \in \mathcal{X}$. The density of real segments is therefore

$$f(x) = (1 - \pi) f_0(x) + \pi \sum_{k=1}^M w_k f_k(x), \quad (12)$$

and we define the *local template intensity*

$$\lambda(x) = \frac{\sum_k w_k f_k(x)}{f_0(x)}, \quad (13)$$

which is the template mass near x relative to how common the shape x is under the null.

Assumption A2 (benchmark validity). The random-walk segments are draws from f_0 , in a comparison set of the same size N as the real block. This holds by construction: the benchmark reshuffles the same pool of returns within the same calendar year, applies the same within-window standardization, and is searched with the same K nearest neighbors (Section 2.3).

Assumption A3 (regularity). f_0 and f are continuous and positive on $\mathcal{X}$, and the nearest-neighbor asymptotics $K \rightarrow \infty$, $K/N \rightarrow 0$ apply as $N \rightarrow \infty$.

Lemma 1 (nearest-neighbor distances estimate local density). *Under A3, for a query point x and a comparison sample of N draws from a density g , the mean distance to the K nearest neighbors satisfies*

$$\bar{d}_K(x) = c_p \left(\frac{K}{N g(x)} \right)^{1/p} (1 + o_p(1)), \quad (14)$$

where c_p is a dimensional constant. Nearest-neighbor distances are therefore inversely related to the local density $g(x)$ (Loftsgaarden and Quesenberry, 1965).

What the PDR identifies

Proposition 1 (PDR is a local density ratio; behavior under the null). *Under A1–A3, applying Lemma 1 to the numerator of the PDR (comparison density f , sample size N) and to its denominator (comparison density f_0 , sample size N) gives*

$$\text{plim } PDR(x) = \left(\frac{f_0(x)}{f(x)} \right)^{1/p} = \left[(1 - \pi) + \pi \lambda(x) \right]^{-1/p}. \quad (15)$$

The constants c_p , K , and N cancel exactly. In particular, if $\pi = 0$ (no recurring structure), then $f = f_0$ and $PDR(x) \rightarrow 1$ for every x . Because the population PDR is then constant, sample PDRs computed on disjoint comparison blocks are independent estimation noise, and the correlation between in-sample and out-of-sample PDR converges to zero.

The empirical counterpart of the null case is the random walk placebo. Its pooled in-sample/out-of-sample correlation is 0.011 (Table 1), compared with 0.308 in real data, and its transition matrices have all entries near 0.20 (Table 2, Panel E). The proposition predicts a population correlation of exactly zero. A finite placebo instead gives a small residual specific to its realization, so the informative comparison is one of order of magnitude, not of statistical significance.

Remark 1 (why the raw neighbor distance is not a recurrence measure). By Lemma 1, the numerator alone satisfies $\bar{d}_n^{real \rightarrow real} \propto (K/(Nf(x)))^{1/p}$. It depends on the total local density f , and hence on the generic commonness $f_0(x)$ of the segment's shape and on the comparison-set size N . Sorting on the numerator therefore sorts jointly on shape commonness, sample density, and recurrence. The ratio in Equation (15) does not depend on $f_0(x)$, N , K , or c_p . It depends on the data-generating process only through $\lambda(x)$, the template mass *relative to* generic commonness. This is the formal version of the discussion in Section 2.4.

Proposition 2 (two-sided detection and compression). *Under A1–A3: (i) $\text{plim } PDR(x) < 1$ if and only if $\lambda(x) > 1$, and the limit is strictly decreasing in $\lambda(x)$: a low PDR identifies regions of trajectory space where real data are denser than chance implies. (ii) In template-free regions ($\lambda(x) \approx 0$), $\text{plim } PDR(x) = (1 - \pi)^{-1/p} > 1$. Because a fraction π of real mass is drawn into template regions, segments far from every template are matched less closely in real data than in the benchmark. A high PDR therefore signals informative uniqueness, not noise. (iii) The exponent $1/p$ compresses deviations from one: for small $\pi(\lambda - 1)$, $\text{plim } PDR(x) \approx 1 - \pi(\lambda(x) - 1)/p$.*

Part (iii) explains why the empirical PDR distribution is tightly concentrated around one, while its *ranking* remains highly informative. Mean PDR ranges only from 0.982 in the lowest quintile to 1.038 in the highest (Table 1, Panel A). With $p = 58$, even a large template intensity moves the level of the ratio only slightly, so small but stable differences in PDR reflect large differences in local density.

Persistence

Proposition 3 (persistence of recurrence). *Suppose the template structure (τ_k, w_k, π) is stable across comparison blocks. Then the population PDR, $\psi(X_n) = [(1 - \pi) + \pi \lambda(X_n)]^{-1/p}$, is a fixed attribute of a segment, and estimated PDRs on disjoint blocks satisfy $\widehat{PDR}_n^{in} = \psi(X_n) + e_n^{in}$ and $\widehat{PDR}_n^{out} = \psi(X_n) + e_n^{out}$ with independent estimation errors. Consequently: (i) $\text{corr}(\widehat{PDR}_n^{in}, \widehat{PDR}_n^{out}) = \text{Var}(\psi)/[\text{Var}(\psi) + \text{Var}(e)] > 0$; (ii) the correlation rises as estimation*

noise falls, that is, as the effective size of the comparison sample grows; (iii) under the null, $\text{Var}(\psi) = 0$ and the correlation is zero; (iv) quintile retention is highest where ψ differs most from its median relative to noise, that is, in the two tails of the λ distribution.

Part (i) is the persistence documented in Table 1. Part (ii) is confirmed directly by the block-length specifications. Lengthening the comparison blocks from one to three years enlarges the comparison sample and raises the correlation from 0.308 to 0.486. In contrast, when the blocks are held fixed at one year, the correlation declines only gently with temporal distance (0.308 to 0.270 at three years and 0.257 at five; Table 1, Panel D). Part (iii) is the placebo. Part (iv) matches the U-shaped diagonal of the transition matrices (Table 2): both extremes persist more than the middle quintiles. The template-free region is the most stable of all, because there $\lambda(x) \approx 0$ and there are almost no template weights to drift. This is consistent with Q5 retention (0.41–0.55) exceeding even Q1 retention (0.27–0.35).

Predictability

Assumption A4 (templates carry return information). Template k is associated with a conditional mean μ_k of the future characteristic-adjusted return of the segments it generates; null draws have conditional mean zero.

Proposition 4 (recurrence and predictability). *Under A1–A4, the expected future adjusted return of a segment at location x is*

$$E[r | x] = \sum_{k=1}^M P(k | x) \mu_k, \quad P(k | x) = \frac{\pi w_k f_k(x)}{f(x)}, \quad (16)$$

and the total posterior template probability $q(x) = \pi \lambda(x) / [(1 - \pi) + \pi \lambda(x)]$ is monotone decreasing in the population PDR. Consequently: (i) averages of the nearest neighbors' realized outcomes (the neighbor T -statistic of Appendix B) estimate the local conditional mean and are predictive out of sample; (ii) the reliability of that prediction increases with $q(x)$ and therefore decreases with PDR, so recurrence strength and predictability are linked; (iii) $E[r | x]$ is a smooth function of location in shape space, so it can be approximated by ridge regression on kernel similarities to a fixed set of landmarks whose coverage includes the template regions. This is the estimator of Section 5; (iv) under the null, $E[r | x] \equiv 0$: there is no pattern-based predictability to find.

Part (ii) corresponds to the decline in sign preservation toward the most novel PDR quintile (Table 3).

Remark 2 (the landmark basis is interchangeable). Proposition 4(iii) requires only that the landmarks' coverage include the template regions; it does not require that the landmarks be selected on recurrence. The information the estimator uses lies in the training pairs of trajectories and realized outcomes. The landmarks only fix the finite-dimensional basis in which the kernel regression is expressed. Any landmark set that covers the support of the data thus spans nearly the same space of smooth functions of location. Moreover, real segments are drawn from the mixture f of Equation (12). A landmark set drawn at random from the real data therefore places mass in template regions in proportion to their density, which is the density structure documented in Section 3. As a result, random real-data landmarks are not a recurrence-free alternative to the recurrence grid. The recurrence–predictability link is tested instead by the landmark-independent implications of Proposition 4(ii): the gradient in Table 3.

Discussion

Remark 3 (misspecification of the null). The cross-stock shuffle removes each stock’s own time-series dependence *and* makes the cross-section homogeneous. Suppose returns show univariate autocorrelation (for example, volatility clustering) or persistent stock-level differences in their return distributions (stocks with systematically different tail, skewness, or jump behavior). Then the empirical benchmark understates f_0 in the affected regions, and the PDR can flag recurrence that reflects only own-stock dependence or distribution type. The block-bootstrap surrogate of Section 3.3 re-estimates the benchmark under a null that keeps most of each stock’s own autocorrelation and its factor exposures. Persistence survives at about three quarters of its baseline strength, which indicates that the template structure is not an artifact of the choice of f_0 .

Remark 4 (economic content). The model says nothing about where templates come from, and this is exactly what makes it a framework rather than a theory. Any mechanism that generates repeated conditional return dynamics, whether behavioral, institutional, or informational, maps into some (τ_k, w_k, μ_k) . The evidence in Section 5.3 (somewhat stronger signal quality in small stocks) suggests that at least part of the template-linked return predictability behaves like mispricing. As discussed there, however, a risk-based explanation cannot be ruled out.

In sum, the framework maps one-to-one onto the empirical design of the paper. Proposition 1 underlies the random walk placebo. Proposition 2 explains why the PDR distribution is tight but informative, and why uniqueness is informative. Proposition 3 delivers the persistence and transition-matrix results of Section 3. Proposition 4 delivers the recurrence–predictability link of Section 4 and the design of the portfolio strategy in Section 5.